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Five-layer systems analysis of *Leishmania* stage differentiation reveals an essential role for protein degradation in parasite development

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eLife Assessment

This **important** study provides a comprehensive multi-omics characterization of *Leishmania donovani* stage differentiation, offering insights into the molecular basis of parasite adaptation across host environments. The authors present **convincing** evidence that stage transitions are not driven by genomic variation but instead rely on coordinated post-transcriptional regulation, including mRNA turnover, translation, and protein degradation. Although experimental validation of these findings and conclusions remains to be completed, the integration of diverse, high-quality datasets establishes a robust resource that will be of broad utility to researchers investigating *Leishmania* biology and life-cycle progression.

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Abstract

Vector-borne, protist parasites have evolved complex developmental programs to adapt to very distinct host environments. How these important pathogens transition between insect and mammalian stages is only poorly understood. Here we investigate stage differentiation in the trypanosomatid parasite *Leishmania* that shows constitutive gene transcription, thus providing a unique model system to assess how development is governed by post-transcriptional mechanisms. Using a five-layer integrative systems analysis (from genome to metabolome), we examined hamster-isolated *Leishmania donovani* amastigotes and culture-derived, insect-stage promastigotes. This approach enabled us to rule out genomic adaptation as a key driver of parasite stage differentiation, while confirming the pivotal role of differential mRNA turnover in stage-specific gene expression. Assessing transcriptomic against proteomic expression changes uncovered an unexpectedly broad dynamic range of stage-regulated changes in protein abundance that only poorly correlated with mRNA levels. This discrepancy correlated with (i) alterations in snoRNA expression and changes in rRNA modification they guide suggesting stage-specific adaptation of the protein translation apparatus that can uncouple mRNA from protein

abundancies, and (ii) differential protein degradation as revealed by quantitative proteomics of parasites treated with the proteasomal inhibitor lactacystin. Lactacystin treatment stalled the transition of spleen-derived amastigotes into promastigotes in culture, further underscoring the role of proteasomal activity in stage differentiation. Integration of our five-layer systems analysis established the first link between *Leishmania* development and the expression of co-regulated genetic networks encompassing mRNA turnover, protein translation, phosphorylation, and degradation. Our findings provide a powerful new resource for research programs that aim to dissect the emergent properties of regulatory networks and feedback loops underlying *Leishmania* stage differentiation, serving as a blueprint for other vector-borne pathogens that rely on disease-associated developmental transitions.

Introduction

During their infectious cycle, many protist and fungal pathogens undergo complex developmental transitions that adapt their biology to different host environments, thus ensuring transmission, persistence or immune evasion. For example, stage differentiation of the malarial parasite *Plasmodium* spp is tightly controlled at transcriptional levels by the dynamic interplay between epigenetic regulators that remodel the chromatin landscape to accommodate stage-specific transcription factors, such as AP2 DNA-binding family members (1, 2). Likewise, the transition from fast-growing tachyzoites to slow-growing bradyzoites during mammalian *Toxoplasma* infection is regulated by the single master TF BFD1 (3). Unlike these apicomplexan parasites, stage-differentiation in the trypanosomatid pathogen *Leishmania* spp is not regulated at the level of transcriptional control, raising the question of how these parasites establish and maintain the various adaptive forms that develop inside their insect and mammalian hosts.

Leishmania spp are protist parasites of humans that represent a global public health problem causing a series of immunopathologies termed the leishmaniasis (4–7). These parasites differentiate into various developmental forms that are adapted for extracellular proliferation inside the sand fly midgut (procyclic promastigotes), transmission from the vector to the mammalian host during uptake of a blood meal (metacyclic promastigotes), and intracellular proliferation inside fully acidified macrophage phagolysosomes (amastigotes) (8). Unlike other eukaryotes, *Leishmania* development is not controlled transcriptionally as protein-coding genes in these parasites lack individual promoters and are constitutively transcribed at all stages involving parasite-specific processes such as polycistronic transcription and trans-splicing (9, 10). Despite the largely constitutive gene transcription, different *Leishmania* stages are characterized by striking morphological, biochemical and metabolic changes (11–16). These are regulated by a plethora of post-transcriptional mechanisms, including differential mRNA turn over through the binding of various proteins and protein complexes to the 5' and 3' untranslated regions (UTRs) (17–22), that can trigger mRNA degradation via deadenylation and decapping (23, 24). Differential gene expression in *Leishmania* can further be regulated at the translational level, for example through (i) selective ribosome recruitment by one of the six isoforms of the *Leishmania* cap-binding protein eIF4E that bind to the 5' UTR, (ii) stabilizing the poly(A) tail via the recruitment of Poly(A) Binding Protein 1 (PABP1) to the 3' UTR promoting translation, or (iii) the differential expression of ribosomal proteins that can increase the landscape of specialized ribosomes (25–34). Finally, differential gene expression in *Leishmania* can also be controlled at the level of protein stability, through stage-specific expression of proteasomal components, autophagy-related genes (ATGs) or lysosomal proteases (35–41).

While each level of gene expression control has been studied individually in different *Leishmania* species, how these different types of regulation are integrated during *Leishmania* stage differentiation remains to be elucidated. Here, we approached this open question by applying a 5-layer, systems level analysis on the two major life cycle stages of *Leishmania donovani*, i.e. hamster-isolated, *bona-fide* amastigotes and derived promastigotes. Our analysis draws a complex picture of the *Leishmania* differentiation process that emerges from complex, co-regulated genetic networks involving mRNA turnover, protein translation, protein phosphorylation and protein degradation. Our data reveal *Leishmania* as an interesting model system to analyze phenotypic

adaptation in the absence of transcriptional control and to dissociate the role of complex regulatory networks in the development of stable, disease-causing life cycle stages in eukaryotic pathogens.

Material & methods

Ethics statement

Work on animals was performed in compliance with French and European regulations on care and protection of laboratory animals (EC Directive 2010/63, French Law 2013-118, February 6th, 2013). All animal experiments were approved by the Ethics Committee and the Animal welfare body of Institut Pasteur and by the Ministère de l'Enseignement Supérieur, de la Recherche et de l'Innovation (projects n°#19683 and #240013).

Animals

Twenty-five female Golden Syrian hamsters (*Mesocricetus auratus* RjHan:AURA, weighting between 50 – 60 g) were purchased from Janvier Laboratories. All animals were handled under specific, pathogen-free conditions in biohazard level 3 animal facilities (A3) accredited by the French Ministry of Agriculture for performing experiments on live rodents (agreement A75-15-01).

Parasites and culture

Leishmania donovani strain 1S2D (MHOM/SD/62/1S-CL2D) was obtained from Henry Murray, Weill Cornell Medical College, New York, USA and maintained by serial passages in hamsters. Anesthetized hamsters were inoculated by intra-cardiac injection of 5×10^7 amastigotes purified from hamster infected spleens as described previously (42). The weight of the animals was recorded over time, and the animals were euthanized by CO₂ asphyxiation before they reached the end-point of infection represented by a 20% loss of body weight. Amastigotes were then recovered from the infected hamster spleens and used for nucleic acid and protein extractions, or differentiated into promastigotes at 26°C in M199 complete medium (M199, 10% FBS, 25 mM HEPES; 100 µM adenine, 2 mM L-glutamine, 10 µg/ml folic acid, 13.7 µM hemin, 4.2 mM NaHCO₃, 1xRPMI1640 vitamins, 8 µM 6-biopterin, 100 units penicillin and 100 µg/ml streptomycin, pH 7.4). Promastigotes, derived from splenic amastigotes, were collected after 2 *in vitro* passages for nucleic acid and protein extractions.

Experimental design

Strains issued from independent experimental evolution assays are identified by number, e.g. amastigote ama1 and promastigote pro1 are the parasites prepared from hamster H1 (see Figures 1A and S1 for details). For comparative analyses, DNA, RNA, proteins and metabolites were extracted from different splenic ama and their matching pro parasites, the latter being collected for extraction during exponential growth phase.

Parasite differentiation and lactacystin treatment

Spleen-derived amastigotes were incubated at 2×10^7 /ml and 26°C in presence or absence of 10 µM of lactacystin (L6785, Sigma) for 3, 6, 18 and 48h in M199 complete medium. At each time point, parasites were collected to control for the presence of paraflagellar rod protein 2 (PFR2) as promastigote differentiation marker. After one *in vitro* passage, amastigote-derived promastigotes were incubated at a concentration of 2×10^7 parasites per ml in presence or absence of 10 µM of lactacystin for 18h. Parasite viability was assessed by FACS analysis after 18h of treatment by propidium iodide (1µg/ml) or YO-PRO-1 (0.2 µM) (# Y3603, Invitrogen) staining as readout using untreated and PFA-treated parasites as controls. Parasites were collected from 25 ml of culture after 18h of treatment for quantitative, label-free proteomics analyses.

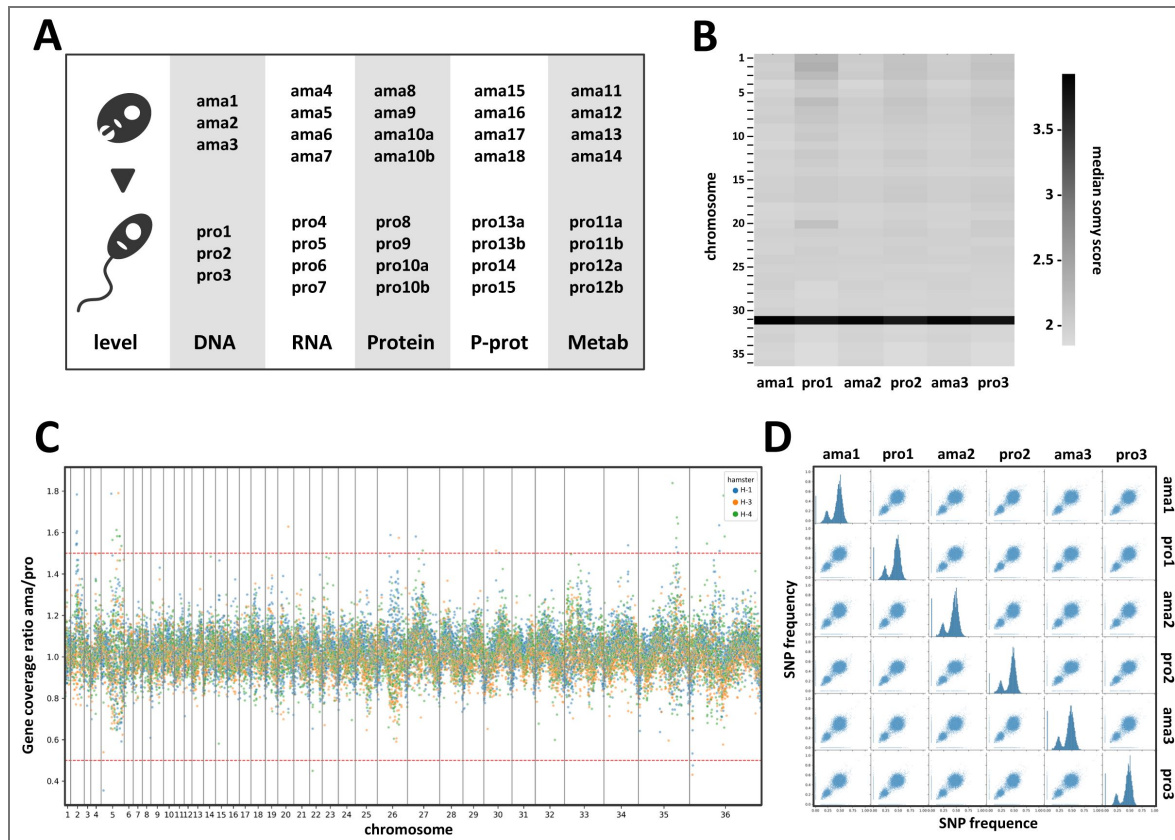


Figure 1. Comparative genomic analysis of *L. donovani* lesion-isolated amastigotes and derived promastigotes.

(A) Schematic overview of the parasite samples used for the various systems level analyses presented in this study. A more detailed overview is shown in Figure S1. (B - D) Genomic analyses. (B) Heatmap showing the somy score of three individual differentiation experiments using hamster-isolated amastigotes (ama1, ama2 and ama3) and their corresponding, culture-derived promastigotes analysed at passage 2 (pro1, pro2 and pro3). Samples and chromosomes are indicated on the x- and y-axis, respectively. The somy scores were calculated as described in Material and Methods and correlate to the grey level according to the shown legend. (C) Ratio plot showing the gene coverage ratio ama versus pro (y-axis) for all genes across the 36 chromosomes (x-axis) as calculated based on median read depth normalized by the somy score. Each color represents one individual differentiation experiment using ama isolates from individual hamsters. The red dotted lines indicate the ratio values corresponding to 1.5 (upper line) and 0.5 (lower line), indicating respectively gain or loss of one gene copy in the ama samples. Fluctuations between 0.5 and 1.5 were not considered significant. (D) Correlation plots representing the individual (histograms on the diagonal) and pairwise (off-diagonal scatterplots) distributions of SNP frequencies for the indicated samples. Of note, only SNPs with a frequency above 10% were plotted. Due to bottle neck events, some low frequency SNPs appear to be unique in one or the other stage as they are either 'lost' (filtered out) or 'gained' (passing the 10% cutoff) between the ama and pro samples. The absence of SNPs at 100% is explained by the heterozygous nature of the Ld15 genome.

Microscopy

Differential interference contrast (DIC) microscopy was applied on hamster-derived amastigotes during *in vitro* differentiation to promastigotes using an Axioplan 2 imaging microscope using a 63x oil immersion objective, the AxioVision Rel.4.8 software and an AxioCam MRm camera (Carl Zeiss). Alternatively, microscopic images of amastigotes in presence or absence of the lactacystin inhibitor were obtained using the EVOS FL microscope (life technologies) at a 20x magnification.

Western blot analysis

Protein extracts were obtained from parasites after 3, 6, 18 and 48h of differentiation from amastigotes to promastigotes in presence or absence of lactacystin. Extracts were loaded on 4-12% SDS-PAGE (NuPage 4-12% Bis-Tris gel NP0321BOX, Invitrogen), separated by electrophoresis and stained with Coomassie blue or transferred onto PVDF membrane. The presence of PFR2 and α -tubulin was revealed using the anti-PFR2 antibody (kindly given by Philippe Bastin), an anti- α -tubulin antibody (clone B-5-1-2, Sigma), the corresponding HRP-conjugated secondary antibodies (Invitrogen) and the SuperSignal West Pico PLUS kit (Thermoscientific). Protein extracts from lactacystin-treated or untreated amastigotes and promastigotes were labelled with Cy5, loaded on 4-12% SDS-PAGE (NuPage 4-12% Bis-Tris gel NP0321BOX, Invitrogen), separated by electrophoresis and transferred onto PVDF membrane. The presence of ubiquitinated proteins was revealed using the mouse anti-ubiquitin monoclonal antibody P4D1 (Abcam), the HRP-conjugated anti-mouse secondary antibody (Invitrogen) and the ECL Prime western blotting detection reagent (Cytiva). Images were acquired using the ImageQuant 800 (Cytiva).

Genome sequencing and data analysis

DNA was prepared from splenic amastigotes and promastigotes at exponential culture phase (three biological replicates). Parasites were centrifuged at 1,600 x g (pro) or 2,000 x g (ama) for 10 min at room temperature. Approximately 1×10^8 promastigotes from logarithmic growth phase were re-suspended in 200 μ l PBS and genomic DNA was purified using DNeasy Blood and Tissue kit from Qiagen and RNase A, according to the manufacturer's instructions. DNA concentrations were measured in duplicate by fluorescence using a Molecular Device fluorescence plate reader (Quant-IT kits, Thermo Fischer Scientific). The quality of DNA was controlled by determining the DNA Integrity Number (DIN) analyzing 20 ng of DNA on a TapeStation 4200 (Agilent). One μ g of genomic DNA was used to prepare a library for whole genome sequencing on an automated platform, using the Illumina "TruSeq DNA PCR-Free Library Preparation Kit", according to the manufacturer's instructions. After normalization and quality control, qualified libraries were sequenced on a HiSeqX5 platform from Illumina (Illumina Inc., CA, USA) at the Centre National de Recherche en Génétique Humaine (CEA, Evry, France), generating paired-ended, 150-bp reads. Sequence quality parameters were assessed throughout the sequencing run. Standard bioinformatics analysis of sequencing data was based on the Illumina pipeline to generate a FASTQ file for each sample.

Genomic DNA reads were aligned to the *L. donovani* Ld1S reference genome (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA396645> [GCA_002243465.1](#)) with BWA mem (version 0.7.12) with the flag -M to mark shorter split hits as secondary. Samtools fixmate, sort, and index (version 1.3) were used to process the alignment files and turn them into bam format [\(43\)](#). RealignerTargetCreator and IndelRealigner from the GATK suite were run to homogenize indels [\(44\)](#). Eventually, PCR and optical duplicates were labeled with Picard MarkDuplicates (version 1.94(1484)) (<https://broadinstitute.github.io/picard/> [\(45\)](#)) using the option "VALIDATION_STRINGENCY=LENIENT". For each read alignment file, Samtools view (version 1.3) and BEDTools genomecov (version 2.25.0) were used to measure the sequencing depth of each nucleotide [\(45\)](#). Samtools was run with options "-q 50 -F 1028" to discard reads with a low map quality score or potential duplicates, while BEDTools genomecov was run with options "-d -split" to compute the coverage of each nucleotide. The coverage of each nucleotide was divided by the median genomic coverage. This normalization is done to account for library size differences. The chromosome sequencing coverage was used to evaluate aneuploidy between EP.1 and LP.1

samples. Then for each sample and for each chromosome, the median sequencing coverage was computed for contiguous windows of 2,500 bases. As previously published (46), the stably disomic chromosome 36 was used to normalize chromosome read depth and to estimate chromosome polysomy levels in each sample. Gene counts were produced using featureCounts (version 1.4.6-p3 (47)) with these parameters: -s 0 -t gene -g gene_id and were normalized according to the median-ratio method.

For the generation of the chromosome median somy score heatmap, mean coverage in 300 bp bins as generated by the GIP pipeline (48) were used to compute somy scores per chromosome by first normalizing bin scores for a sample by their median across all genome (to obtain comparable values between samples), multiplying by two (to scale somy values to the default diploid state assumed for most of the genome), and then taking the median across bins belonging to a given chromosome (see also <https://gip.readthedocs.io/en/latest/giptools/karyotype.html> (49)). These chromosome scores were then displayed on a sample x chromosome heatmap, excluding the “maxi” chromosome. This was done using the Pandas 1.4.2 (49, 50) Pandas-dev/pandas: Pandas 1.2.4, <https://zenodo.org/records/4681666> (50), Matplotlib 3.5.1 (51) and Seaborn 0.11.2 (52) Python libraries.

For the assessment of gene coverage ratios, normalized mean coverages per gene are reported by GIP (version 1.1.0 (48)), as the mean coverage of the gene divided by the median coverage of the chromosome containing the gene. For a given gene, and a given amastigote - promastigote pair, a gene coverage ratio was computed adding 0.1 to the gene coverages of the individual samples, in order to avoid zero division errors. The genes were sorted by genomic coordinate and assigned successive integers to create a “genomic index”. Gene coverage ratios were then plotted along this genomic index, using the same Python libraries as above.

For the SNP analysis, alternative (Alt) allele frequencies for SNPs were retrieved from the output of the SNV giptools module (provided along with the GIP pipeline). Euclidean distances between samples in the SNP allele frequencies space were computed and used to build a dendrogram using the scipy.cluster.hierarchy Python module (SciPy version: 1.8.0). The clustering was obtained using the UPGMA method.

RNA-seq analysis

Total RNA was extracted from splenic amastigotes and promastigotes in exponential culture phase (4 biological replicates). Amastigotes and promastigotes were centrifuged respectively at 2,000 x g or 1,600 x g for 10 min at 4°C, and re-suspended in the lysis buffer supplied with the Qiagen RNeasy Plus kit. The samples were stored at -80°C and RNA extractions were performed according to the manufacturer’s instructions, including a DNase treatment. RNA integrity was validated using the Agilent Bioanalyzer. DNase-treated RNA extracts were used for library preparation. Libraries were constructed using an Illumina Stranded mRNA Prep (Illumina, USA) following the supplier’s recommendations. RNA sequencing was performed at the Biomix Center (Institut Pasteur, Paris, France) on the Illumina NextSeq 2000 platform for a target of 40M paired-end reads per sample.

The RNA-seq analysis was performed using the Sequana RNA-seq pipeline (version 0.20.1, https://github.com/sequana/sequana_rnaseq (53)) from the Seauqng project (53). The pipeline workflow manager is Snakemake 7.3.2 (54). The pipeline was executed with default parameters and bioinformatics software are available as reproducible containers from the Damona project (<https://github.com/cokelaer/damona> (55)). Reads were mapped to the *Leishmania donovani* reference using bowtie2 2.4.5 (55). Gene-level quantification was conducted using FeatureCounts 2.0.1 (47) assigning reads to genomic features based on the gene annotation and gene_id attribute, while accounting for strand-specificity information. Differential expression analysis was performed using DESeq2 v1.24.0 (56) scripts available in the Sequana library. Statistical testing identified differentially expressed genes by comparing amastigote and promastigote sample groups, with significance determined using Benjamini-Hochberg adjusted p-values (false discovery rate FDR < 0.05).

To specifically assess the differential expression of snoRNA, RNA-seq libraries from promastigote and amastigote stages were mapped to the *Ld1S* genome, and snoRNA read counts were obtained using bedtools multiBamCov. For snoRNAs with multiple copies, read counts were summed per sample to generate a single expression value per snoRNA family. DESeq2 was then used to perform differential expression analysis between amastigote and promastigote stages based on the summed snoRNA counts.

Label-free, quantitative total proteome analyses

Untreated amastigotes (ama and ama-18h) and promastigotes (pro), and lactacystin-treated parasites (ama lacta and pro lacta) were recovered from four replicates (Figure 1A [↗](#) and S1 [↗](#)) and washed three times with cold PBS at 2,000 x *g* or 1,600 x *g* for 10 min at 4°C. Parasite lysates were prepared in eFASP lysis buffer (4% SDS / 0.2 % DCA / 50mM TCEP / 50 mM ammonium bicarbonate buffer pH 8) ([61](#)). For detailed protocol see supplementary data file. MS scans were acquired at a resolution of 70,000 and MS/MS scans (fixed first mass 100 *m/z*) at a resolution of 17,500. The AGC target and maximum injection time for the survey scans and the MS/MS scans were set to 3 x 10⁶ for 20 ms and 10⁶ for 60 ms, respectively. An automatic selection of the 10 most intense precursor ions was activated (Top 10) with a 40 s dynamic exclusion. The isolation window was set to 1.6 *m/z* and normalized collision energy fixed to 28 for HCD fragmentation. We used a minimum AGC target of 10⁴ corresponding to an intensity threshold of 1.7 x 10⁵. Unassigned precursor ion charge states as well as 1, 7, 8 and >8 charged states were rejected and match was disabled.

Label-free quantitative phosphoproteome analyses

Amastigote (ama) and promastigote (pro) parasites from four biological replicates were recovered and washed three times by centrifugation in cold M199 at respectively 2,000 x *g* or 1,600 x *g* for 10 min at 4°C. Samples were incubated for 10 min at 4°C in lysis buffer (1ml per 1.5x10⁹ parasites) consisting of 8 M urea, 50 mM Tris, supplemented with a protease inhibitor cocktail (cOmplete from Roche) and a phosphatase inhibitor cocktail (PhosStop from Roche). Following sonication for 5 min using a sequence of 10 s pulse and 20 s pause, the lysates were centrifuged 15 min at 14,000 x *g* and 4°C, and the supernatant was collected and stored at -80°C until use. Proteins were quantified by RC DC protein assay (Bio-Rad) and adjusted to 1.3 µg.µl⁻¹ in lysis buffer. For detailed protocol see supplementary data file. Phosphopeptide enrichment was carried out as described in ([62](#)) and detailed in supplementary data. All analyses were performed on a Q Exactive HF Mass Spectrometer (Thermo Fisher Scientific) coupled with a Proxeon EASY-nLC 1000 (Thermo Fisher Scientific) as detailed in supplementary data. Details for the proteomics and phosphoproteomics analyses following the data acquisition are available in supplementary data. Results are presented in table 6 and 10.

Phosphopeptides were selected for Gene Ontology enrichment analyses if (i) they were exclusively identified in one of the two stages, (ii) they showed significant, stage-specific changes in phosphopeptide abundance (fold change ≥ 2, adj. p-value < 0.01) even if the corresponding protein was not detected in the total proteome analysis, and (iii) they showed a significant increase in relative phosphorylation normalized to protein abundance (p-value < 0.05) as calculated by the ratio ‘change in phosphopeptide abundance’ vs ‘change in protein abundance’ using a cut off of fold changes ≥ 2 and adj. p-value < 0.01 for both analyses (see table 11).

Metabolomic analysis

The sample extraction was performed as described previously ([63](#)). Briefly, 10⁸ cells were used per each 200 µL sample (4 replicates/stage). Firstly, cells were rapidly quenched in a dry ice/ethanol bath to 4°C, then centrifuged, washed with 1 x PBS, and resuspended in the extraction solvent (chloroform:methanol:water, 1:3:1). After vigorous shaking at 4°C for 1 h, extracts were centrifuged (16,000 x *g*, 4°C, 10 min) and the supernatants collected and stored at -80°C until the analysis. LC-MS analyses were performed using separation on 150 x 4.6 mm, 5 mm ZIC-pHILIC (Merck) on UltiMate 3000 RSLC (Thermo Scientific) followed by mass detection on an Orbitrap

Exactive mass spectrometer (Thermo Fisher) at Glasgow Polyomics. Analyses were performed in positive and negative polarity switching mode, using 10 μ l injection volume and a flow rate of 300 μ l/min over 26 min on the column maintained at 30°C, as follows: 0 to 20 min 20%-to 80% solution A, 15 to 17 min 95% solution A, 17 to 26 min 20% solution A where solution A is 20 mM ammonium carbonate in water and solution B is acetonitrile). The samples were run alongside 249 authentic standards at 10 μ M each. Mass spectrometry data was processed using Mzmatch (64) and Ideom (63) software. Unique signals were extracted using the centwave algorithm and matched across biological replicates based on mass-to-charge ratio and retention time. These grouped peaks were then filtered based on relative standard deviation and combined into a single file. The combined sets were then filtered on signal-to-noise score, minimum intensity and minimum detections. The final peak set was then gap-filled and converted to text for use Putative metabolite identification corresponds for the most part to Metabolite Standards Initiative (MSI) level 2 (mass only, and thus only considered a tentative annotation), whereas metabolites matching in retention time to an included standard correspond to level 1 (considered likely an accurate annotation). Peaks having an area with root squared deviation across pooled samples > 50% were excluded, as were those with a retention time < 4 min (due to poor resolution).

Localizing rRNA modification on *Leishmania* cryo-EM ribosome structure

The cryo-EM atomic model of *Leishmania major* 80S ribosomes bound to mRNA and all three tRNAs (PDB: 8RXH) was used to project rRNA modifications of the ribosome. Figures were generated using UCSF Chimera-X software (65).

Systems-level analyses

A lists of *L. donovani* GO terms (66) was built in house. The Biological Networks Gene Ontology tool (BiNGO) plugin of the Cytoscape software package (version 3.8.2) was used to map and visualize functional enrichments in each data set based on the GO hierarchy. A Benjamini & Hochberg false discovery rate with a significance level of 0.05 was applied. Cluster efficiency represents the percentage of genes for a given GO term compared to the total number of genes with any GO annotation in the considered set of genes. Enrichment score corresponds to the percentage of genes for a given GO term compared to all the genes sharing the same GO term in the genome. Word Cloud of GO enrichment analysis limited to GO Slim terms and a threshold of p-value < 0.05 for the category Biological Process was performed on TriTrypdb (<https://tritrypdb.org>) using the *L. donovani* LdBPK orthologs for all genes that showed statistically significant expression changes (p-value < 0.01) at both RNA or protein levels.

For the phosphoproteomic work, phosphopeptides were selected for Gene Ontology enrichment analyses if (i) they were exclusively identified in one of the two stages, (ii) they showed significant, stage-specific changes in phosphopeptide abundance (fold change ≥ 2 , adj. p-value < 0.01) even if the corresponding protein was not detected in the total proteome analysis, and (iii) they showed a significant increase in relative phosphorylation normalized to protein abundance (p-value < 0.05) as calculated by the ratio ‘change in phosphopeptide abundance’ vs ‘change in protein abundance’ using a cut off of fold changes ≥ 2 and adj. p-value < 0.01 for both analyses (see table 11).

Functional enrichment networks were built with the STRING plugin of the Cytoscape software package (version 3.8.2) using the *L. infantum* orthologs, and the full STRING network with a confidence score cutoff of 0.4. A false discovery rate with a significance level of 0.05 was applied. The “ClusterProfiler” package (version 4.2.2) of R was used for KEGG (Kyoto encyclopedia of genes and genomes) gene set enrichment analysis (GSEA) and for data mapping on metabolic pathways available in the KEGG database. Results were visualized using the “pathview” packages (version 1.34.0).

Results

***Leishmania* stage differentiation occurs independently from changes in gene dosage**

We carried out an in-depth, systems analysis of independent *L. donovani* amastigote preparations isolated from infected hamster spleens and their culture-derived promastigotes at *in vitro* passage 2, with the aim of revealing mechanisms controlling stage-specific differential expression across five quantifiable information levels, by comparing the genome, transcriptome, proteome, metabolome and phospho-proteome of the two life-cycle stages (Figure 1A and Figure S1). We first investigated the possible role of genome instability in stage-differentiation, given the very well documented propensity of *L. donovani* to establish and tolerate chromosome and gene copy number variations in culture and during sand fly or hamster infection (46, 66, 70).

Applying our computational pipeline termed ‘GIP’ (48) to the genome sequences of three independent pairs of amastigotes and promastigotes confirmed the largely disomic state of the *Leishmania* genome at both stages, except for the known constitutive tetrasomy of chr 31 (Figure 1B and Figure S2A, Table 1). We did not detect any major changes at karyotypic levels following promastigote differentiation, confirming our previous observations that stage differentiation occurs independent of karyotypic adaptation (46, 70). Nevertheless, a clear pattern emerged for promastigotes that showed a slight but reproducible increase in median coverage for chr 1, 2, 3, 4, 6 and 20, suggesting the emergence of mosaic aneuploidy for these chromosomes early during culture adaptation. However, these minor signals are not predictive for the karyotypic changes characteristic of culture-adapted Ld1S parasites (i.e. chr 5, 26, 33, (46, 70)). Likewise, differentiation of promastigotes was not associated with any major gene copy number variation at early passage 2 (Figure 1C, Table 2). Even though a number of genes showed increased (e.g. Ld1S_300477100 encoding an intraflagellar transport (IFT) protein) or decreased (e.g. Ld1S_220235700 encoding a hypothetical protein) read depth in individual pro strains, none of these changes converged across all pro samples. In the absence of convergent selection, it is impossible to distinguish if these gene CNVs provide some strain-specific advantage or are merely the result of random genetic drift. Plotting the normalized read depth ratios between amastigotes and promastigotes revealed an intriguing, undulating pattern that was highly reproducible across the three independent experiments and has been previously associated with nascent DNA produced during the replicative S-phase of the cell cycle (71–73). Finally, differentiation did not affect the SNP frequency distribution (Figure S2B, Table 3), which resulted in a largely diagonal pattern when plotting amastigote against promastigote SNP frequencies (Figure 1D).

In conclusion, unlike long-term culture adaptation, which drives important karyotypic changes (46, 74) that can affect transcript- and protein abundance levels as demonstrated by a recent 4-layer systems analysis (75) *in vitro* differentiation of splenic amastigotes into promastigotes was not associated with significant changes in chromosome or gene copy number, ruling out gene dosage effects as the source of stage-specific expression differences. Thus, *L. donovani* promastigote differentiation *in vitro* is independent of genomic adaptation, confirming our previous results obtained with sand fly-derived, promastigote parasites (66, 76).

Comparative RNA-seq analysis reveals stage-specific, co-regulated gene clusters that are controlled at post-transcriptional levels

We next analyzed stage-specific expression changes applying RNA-seq analyses on four amastigote and derived promastigote strains. In contrast to the relatively stable genome infrastructure maintained between stages, we observed substantial, stage-specific differences in transcript abundance. From 10402 detected transcripts in both stages, 6478 showed statistically significant differences (adj. p-value < 0.01, base mean reads ≥ 10 reads), including 1234 and 1027 transcripts with a 2-fold or higher increase in abundance in amastigotes and promastigotes, respectively (Table 4). Considering the lack of promoter-driven expression control of individual, protein coding genes in *Leishmania* and the absence of stage-specific gene dosage effects (see Figure 1E), the

observed differences in amastigote and promastigote transcript abundances are most likely attributable to stage-specific mRNA turnover. Interestingly, non-coding (nc) RNAs account for 64.7% of the differentially expressed transcripts at the amastigote stage (Figures 2A [↗](#)).

Analyzing the stage-specific RNA-seq data sets identified clusters of co-regulated transcripts that share a common annotation (Figure 2B [↗](#), Table 4). In promastigotes, we observed increased abundance for 66 transcripts annotated for the interconnected GO terms ‘microtubule based process’, ‘cytoskeleton organization’ and ‘ciliary cell motility’, confirming the post-transcriptional co-regulation of flagellar biosynthesis as previously shown in *L. mexicana* (77) (Figure 2B [↗](#) left panel, Table 4, sheet G). Other co-regulated, functional gene sets are defined by the GO terms ‘oxidation reduction’ (46 transcripts), ‘glycolysis’ (15 transcripts), ‘post-transcriptional regulation of gene expression’ (46 transcripts) and ‘translational elongation’ (28 transcripts), further supporting translational control as a regulatory step in stage differentiation.

In amastigotes, a major co-regulated gene cluster is represented by 77 amastin transcripts encoded on 5 different chromosomes that show increased abundance in amastigotes, confirming the stage-specific, post-transcriptional regulation of this gene family as previously described (19, 78, 79) (Table 4, sheet C, E and F). Another functional gene cluster showing increased abundance in amastigotes was associated with protein translation, with GO enrichment observed for the terms ‘ribosome biogenesis’ (26 transcripts), ‘ncRNA processing’ (22 transcripts), including 17 transcripts annotated for the GO term ‘rRNA processing’, and 6 transcripts for ‘tRNA processing’ (Figure 2B [↗](#), middle and right panel, Table 4, sheet E and F).

These data suggest ribosome and epi-transcriptomic regulation as possible central components in *Leishmania* stage differentiation. Indeed, analyzing poly-A transcriptome libraries of amastigotes and promastigotes we observed significant differences in stage-specific expression of 110 pre-snoRNAs (Table 5, Figure 2C [↗](#)). These data fit with the expression of a series of methyl- and pseudouridyl-transferases at this stage (see Table 6, sheets A and E). Stage-specific expression of some of these snoRNAs correlated with stage-specific changes previously observed in Pseudouridine and 2'-O-Methylation rRNA modifications (34). The positions of these modifications and the corresponding snoRNAs known to affect ribosome translational activity are indicated on the high resolution cryo-EM structure of the *Leishmania* ribosome (Figure 2D [↗](#)).

In conclusion, our RNAseq data reveal a series of functionally related, stage-specific gene clusters whose mRNA abundance is regulated in a coordinated manner at post-transcriptional level. Whether this co-regulation is governed by stage-specific mRNA decay mechanisms acting on 5' and 3' UTRs or by stage-specific mRNA stabilization via RNA-binding proteins or mRNA modification remains to be established.

Quantitative proteome analysis reveals co-regulated gene clusters that are independent of stage-specific mRNA turnover

We next employed quantitative proteomic analysis to assess how the different co-regulated gene clusters observed in our RNAseq analyses translate into stage-specific proteomes. Label-free quantitative proteomic analysis of 4 replicates of amastigotes and derived promastigotes identified over 4000 proteins, including 1987 differentially expressed proteins (adjusted p-value < 0.01), and 1534 that were exclusively detected in either ama or pro (Figure 3A [↗](#) left panel, Table 6). Considering the 822 proteins that are specific or more abundant in amastigotes (fold change ≥ 2), functional enrichment was mainly observed for the GO terms ‘oxidation reduction’ (61 proteins), ‘vesicle mediated transport’ (23 proteins) and ‘phosphorylation’ (41 proteins) (Figure 3A [↗](#), middle panel, Table 6, sheets B and D). Considering the 1637 proteins that are specific or more abundant in promastigotes (fold change ≥ 2), functional enrichment was observed for the GO terms ‘post-transcriptional regulation of gene expression’ (71), ‘translation’ (69 proteins), ‘ribosome biogenesis’ (21 proteins) and ‘ciliary cell motility’ (42 proteins) (Figure 3A [↗](#), right panel, Table 6, sheet C and E).

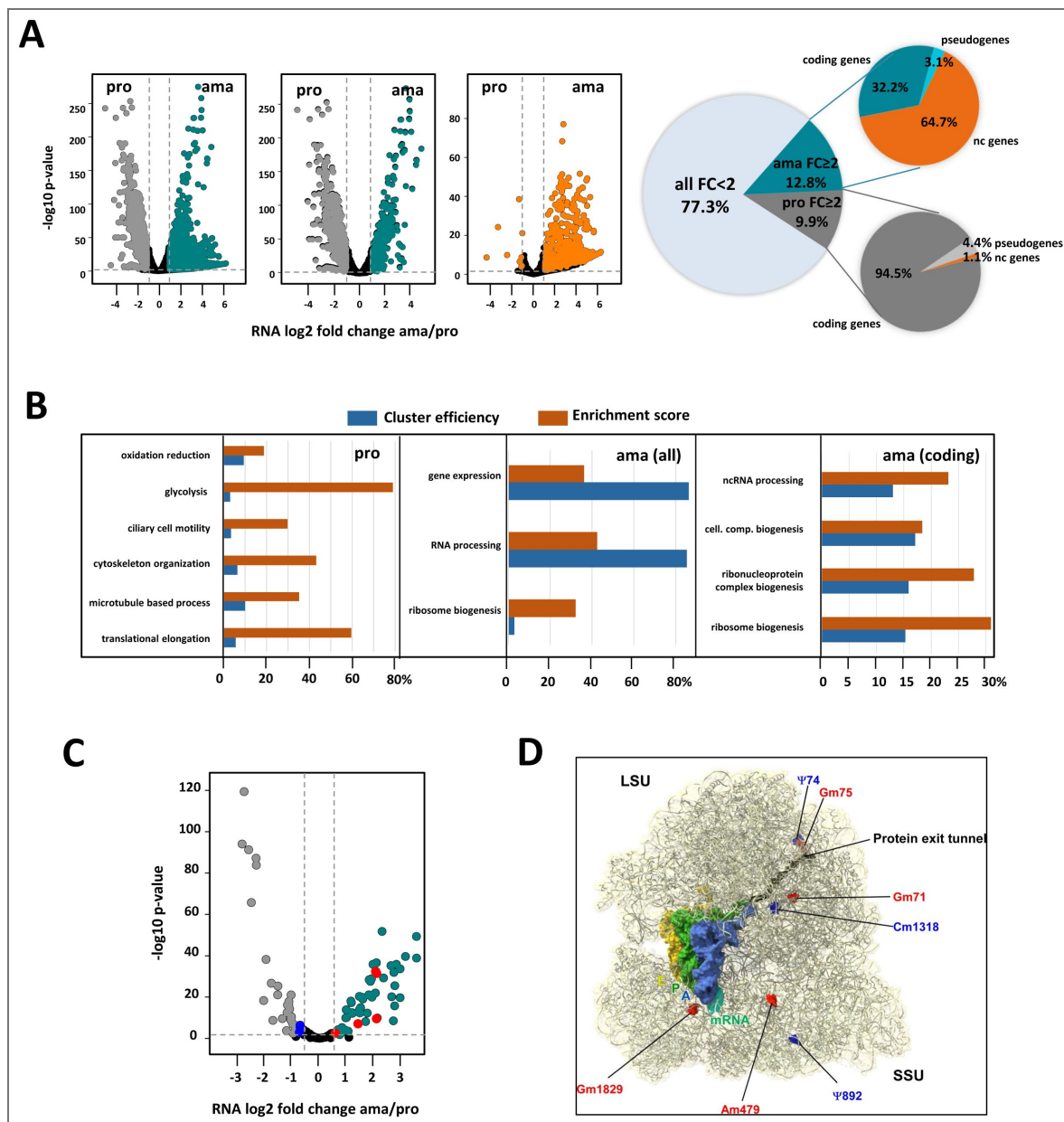


Figure 2. Stage-specific transcript profiling.

(A) Volcano plots (left panels) showing differential transcripts abundance between ama and pro samples as assessed by RNA-seq analysis. The dotted lines indicate fold change (FC) = 2 (vertical line) and p-value = 0.01 (horizontal line). Transcripts with FC < 2 or adjusted p-value > 0.01 are represented by black dots. Transcripts with significant increased abundance FC ≥ 2 and adjusted p-value < 0.01 in ama and pro are indicated respectively in dark cyan and dark grey for all transcripts (left panel) or transcripts of only coding genes (middle panel). Transcripts for non-coding genes with significant increased abundance FC ≥ 2 and adjusted p-value < 0.01 in ama and pro are indicated in the right volcano plot (orange dots). The pie chart (right panel) depicts the percentage of transcripts for each of the categories. (B) GO term enrichment analysis of these transcripts for the category 'biological process' was performed, and 'cluster efficiency' (blue) and 'enrichment score' (orange) were plotted for pro (left panel) and ama datasets (all transcripts, middle panel; transcripts for coding genes, right panel). The graphs show some of the most significantly enriched GO terms associated with a Benjamini and Hochberg (BH) p-value < 0.05 that were identified using the BiNGO plug-in in Cytoscape. (C) Volcano plot corresponding to the differential expression of snoRNA between pro and ama stages. Cyan dots (ama) and grey dots (pro) represent signals with FC ≥ 2 and adj. p-value < 0.05. Black dots represent non-significant changes. (D) Localization of differentially modified rRNA sites on the *Leishmania* ribosome. The RNA modifications are indicated as space filling in the *Leishmania* cryo-EM structure (PDB: 8RXH). The identity of the RNA modification is defined by the label (Ψ, pseudouridine; Gm and Cm refer to guanine and cytosine methylation, respectively; numbers indicate residues). Upregulated and downregulated RNA modifications in amastigotes according to Rajan et al (34) are indicated in red and blue, respectively.

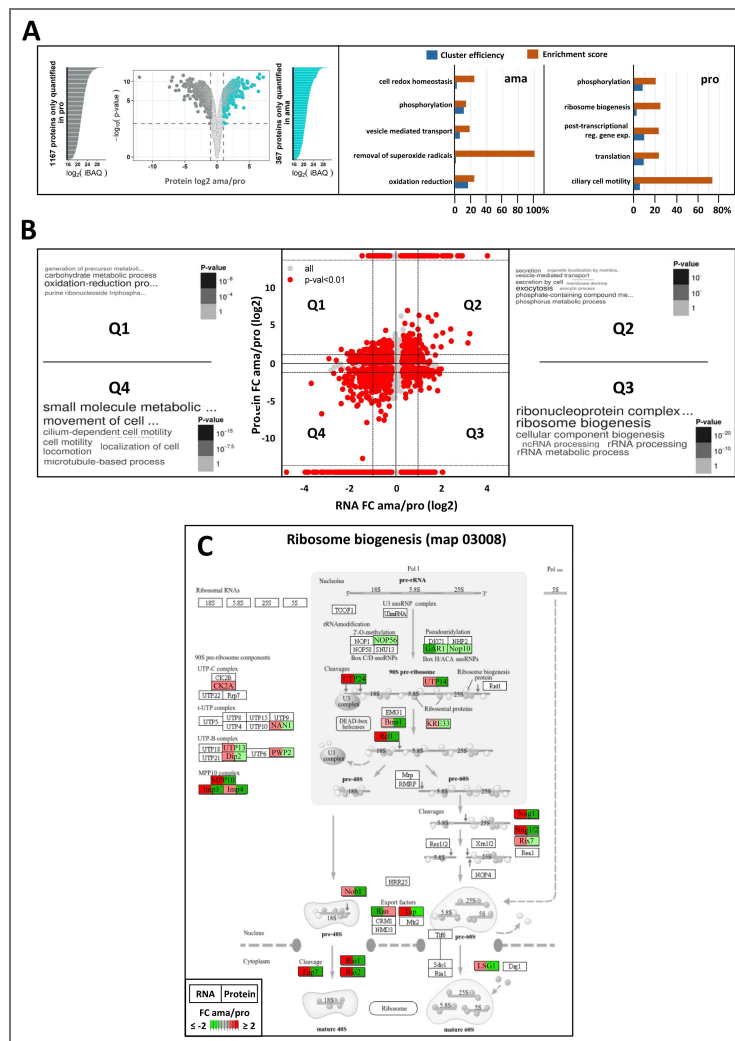


Figure 3. Systems analysis of the stage-specific transcriptome and proteome.

(A) Volcano plot showing differential protein abundance between four ama and pro samples as assessed by label-free, quantitative proteomics analysis. The dotted lines indicate fold change (FC) = 2 (vertical line) and false discovery rate (FDR) = 0.01 (horizontal line). Proteins with FC < 2 or FDR > 0.01 are represented by light grey dots. Proteins that were reproducibly detected in only one of the two stages (considered unique) are represented by the lateral histograms plotting their relative abundance according to the iBAQ value. These proteins together with proteins showing differential abundance of FC ≥ 2 and adjusted p-value < 0.01 are indicated for ama in dark cyan and for pro in dark grey (see Table 6). GO term enrichment analysis for the category 'biological process' was performed, and the resulting values for 'cluster efficiency' (blue) and 'enrichment score' (orange) were plotted for ama (middle panel) and pro (right panel). The graphs show the main GO terms identified using the BiNGO plug-in in Cytoscape and associated with a BH p-value < 0.05. (B) Double ratio plot showing the log2 fold change between ama and pro in transcript (x-axis) and protein (y-axis) abundances. Red dots correspond to changes in both transcript and protein abundance defined either by significant differential abundance at both levels (p-value < 0.01), or by significant RNA changes (p-value < 0.01) corresponding to proteins detected in only one of the two stages. The dotted lines indicate fold changes (FC) = 2. The infinite symbol (∞) represents proteins that are unique in the ama (upper signals) or pro (lower signals) datasets. Word cloud enrichment performed with the *L. donovani* LdBPK orthologs is presented for each of the 4 quadrants, Q1 to Q4 and a detailed GO enrichment analysis is given in Table 7. The font size is proportional to the number of genes per GO term and the grey scale refers to the p-value calculated for each GO term. (C) KEGG map for ribosome biogenesis. Each gene associated with expression changes indicated in Q1 to Q4 (see Figure 2C) and annotated with the GO term 'ribosome biogenesis' has been projected on the ribosome biogenesis KEGG map. Each gene is represented by a square with the left segment showing fold changes between ama and pro at transcript and the right segment showing fold changes at the protein level, with the differential abundance indicated by the color and its intensity (red, increase; green, decrease). Only genes quantified in both transcriptome and proteome analyses and associated with an adjusted p-value < 0.01 were considered.

The metabolomic analysis of four independent amastigote and promastigote samples (Figure S4A) supported our proteomic data. Promastigotes showed increased amino acid biosynthetic activity that matched the changes in protein abundance observed for this pathway, while the amastigote metabolome indicates a shift from glycolytic to TCA cycle-dependent energy production that was previously described (12, 14, 80) (Figures S4B-D), with a notable increase in metabolites involved in butanoate metabolism indicative of a shift between carbohydrate use as a primary energy source to utilization of fatty acid oxidation. The amastigote expression profile and metabolome further indicate the stage-specific production of the storage-carbohydrate mannogen, thus corroborating previous observations (81) (Figure S4E).

Surprisingly, only few enriched GO terms matched between transcriptomics and proteomics datasets, suggesting additional layers of regulation between the transcriptional and translational levels of gene expression. To investigate this aspect on a genome-wide level, we next plotted the simple numerical values we obtained when calculating the ratios of expression changes between ama and pro observed in the transcriptomics and proteomics datasets (Figure 3B, Table 7). Only expression changes were considered that either showed statistically significant differential abundance at both RNA and protein levels ($p < 0.01$), or showed significant RNA changes ($p < 0.01$) with the corresponding protein being detected in only one of the two stages. These latter proteins are identified by signals that were arbitrarily placed at the upper (detected in ama) or the lower (detected in pro) parts of the graph. Whether these proteins just escape detection due to low expression or are truly not expressed remains to be established. The double ratio analysis of the resulting 2349 gene products defined four quadrants: Quadrants 2 (Q2) and 4 (Q4) correspond to 430 and 1042 genes (Table 7, sheet E and I), respectively, for which stage-specific expression changes correlate between protein and RNA levels, suggesting that the abundance of these proteins may be mainly regulated by mRNA turn-over. Q2 (mRNA and protein up in ama) includes the GO terms ‘vesicle mediated transport’ (17 genes) and ‘lipid metabolic process’ (21 genes) (Table 7, sheet F), while Q4 (mRNA and protein up in pro) includes the GO term ‘cell motility’ (41 genes) (Table 7, sheet J).

Counter-correlations between RNA and protein abundances (p -value < 0.01) were observed for 463 and 414 genes, respectively falling into quadrants Q1 (RNA down and protein up in ama) and Q3 (RNA down and protein up in pro) (Figure 3B, Table 7, sheets C and G). The most important difference was observed for 35 genes annotated for the GO term ‘ribosome biogenesis’ (Figure 3C, Table 7, sheet H). While this GO term defined a positively regulated gene cluster at mRNA level in amastigotes (see Figure 2B), the same genes show reduced expression at this stage at protein level (Figure 3C). This lack of correlation may be explained by stage-specific changes in mRNA translation efficiency.

In conclusion, while many stage-specific changes of protein abundance are strictly regulated by similar changes at mRNA level, others show discrepancies in stage-specific abundances between transcript and protein levels, revealing regulatory mechanisms that act independent of mRNA stability, for example at the level of translational control or differential protein stability.

Stage-specific protein turn-over is required for parasite differentiation

The discrepancies we observed in a sub-set of genes between changes in mRNA and protein abundance during the amastigote-to-promastigote transition (see Figure 3B [↗](#)) primed us to investigate the role of proteasomal protein degradation in *Leishmania* differentiation. We chose the highly specific and irreversible proteasome inhibitor lactacystin ([82](#), [83](#)) over the trypanosomatid-specific, reversible drug candidate LXE408 ([84](#)) as the latter's potent cytotoxicity can confound direct effects on protein turnover with secondary consequences of cell death, limiting its utility for dissecting proteasome function in living parasites. Comparative, quantitative proteomics analyses were performed with (i) spleen-derived amastigotes (ama), (ii) amastigotes after the first 18h of differentiation in the absence (control, ama-18h) and the presence of lactacystin (ama-lacta), and (iii) promastigotes in culture without (pro) and after 18h of treatment with lactacystin (pro-lacta) (Table 8, Figure 4A [↗](#) and S5).

This data set allowed us important insight into processes of early differentiation, revealing increased abundance in ama-18h compared to ama for 694 proteins (fold change ≥ 2 , adjusted p-value < 0.01) (Table 8b, sheet B) with GO enrichment observed for the terms 'ribosome biogenesis', 'RNA methylation', 'rRNA processing', 'gene expression', and 'post-transcriptional regulation of gene expression', the latter including 6 genes encoding for pseudouridine transferases known to modify ribosomal (r) RNA (Figures S5A and S6) (Table 8b, sheets D to H). These data further sustain a role of epitranscriptomic regulation and stage-adapted ribosomes as key processes that may initiate *Leishmania* promastigote differentiation, at least in culture ([31](#), [34](#), [66](#)).

Lactacystin treatment further uncovered an essential regulatory role of proteasomal protein degradation in *L. donovani* development, which builds on previous published observations ([85](#)). While the presence of the inhibitor neither affected parasite viability (Figure S7A) nor morphology (data not shown and Figure S7B) or promastigote growth (as previously shown by Silva-Jardim et al ([85](#))), lactacystin treatment abrogated the amastigote-to-promastigote developmental transition as judged by the persistence of an amastigote-like morphology (i.e. oval cell shape, retracted flagellum), and the absence of PFR2 expression characteristic of promastigotes (Figures 4B and C [↗](#)). In contrast, untreated amastigotes spontaneously converted to flagellate promastigotes after 48h in culture (Figure S7D). Proteasome inhibition by lactacystin blocked amastigote-to-promastigote differentiation without inducing rapid global accumulation of ubiquitinated proteins (Figure S7C), which are observed only at later time points and correlated with differentiation-associated remodeling rather than acute proteasome inhibition.

Comparing the proteomics signatures of lactacystin-treated samples and untreated controls (Figure S5) allowed us to reveal the proteasomal targets, whose rescue correlated with the observed abrogation of differentiation. This analysis is complicated by the fact that amastigotes will not only respond to the 18h treatment but also initiate their differentiation into promastigotes during this period, at least in the untreated control sample. To distinguish between changes caused by lactacystin treatment and changes due to the differentiation process, we plotted the ratio of ama/ama-lacta against the ratio ama-18h/ama-lacta (Figure 4D [↗](#)), which defined one quadrant (Q3) representing 129 proteins whose increased abundance in ama-lacta was exclusively due to rescue from degradation (Table 9, sheet A). Applying STRING and GO analyses on these proteins, some of the most significantly enrichment terms included 'protein kinase', 'cellular response to stimulus', and 'cell cycle' (Figure S8 lower panel, Table 9, sheet B and G).

Proteomics analysis of lactacystin-treated promastigotes (pro-lacta) revealed the rescue of 289 proteins from degradation when compared to untreated control (pro) (Table 9, sheet C). Comparison of rescued proteins across both ama and pro stages revealed a series of intriguing findings: First, most of these proteins were rescued in a stage-specific manner, confirming differential protein degradation as an important regulatory component in the development and maintenance of *Leishmania* life cycle stages as suggested by previous reports ([37](#), [86–92](#)). Second, just like the proteins rescued from proteasomal degradation in ama-lacta, the pro-lacta data set too showed GO enrichment for the terms 'protein kinase' and 'cellular response to stimulus', involving distinct sets of proteins (Figure S8, upper panel) (Table 9, sheets D, E and G). Thus, *Leishmania* differentiation correlates with the expression of complex signaling networks that are established in a stage-specific manner. Third, when projecting the lactacystin-rescued proteins to the

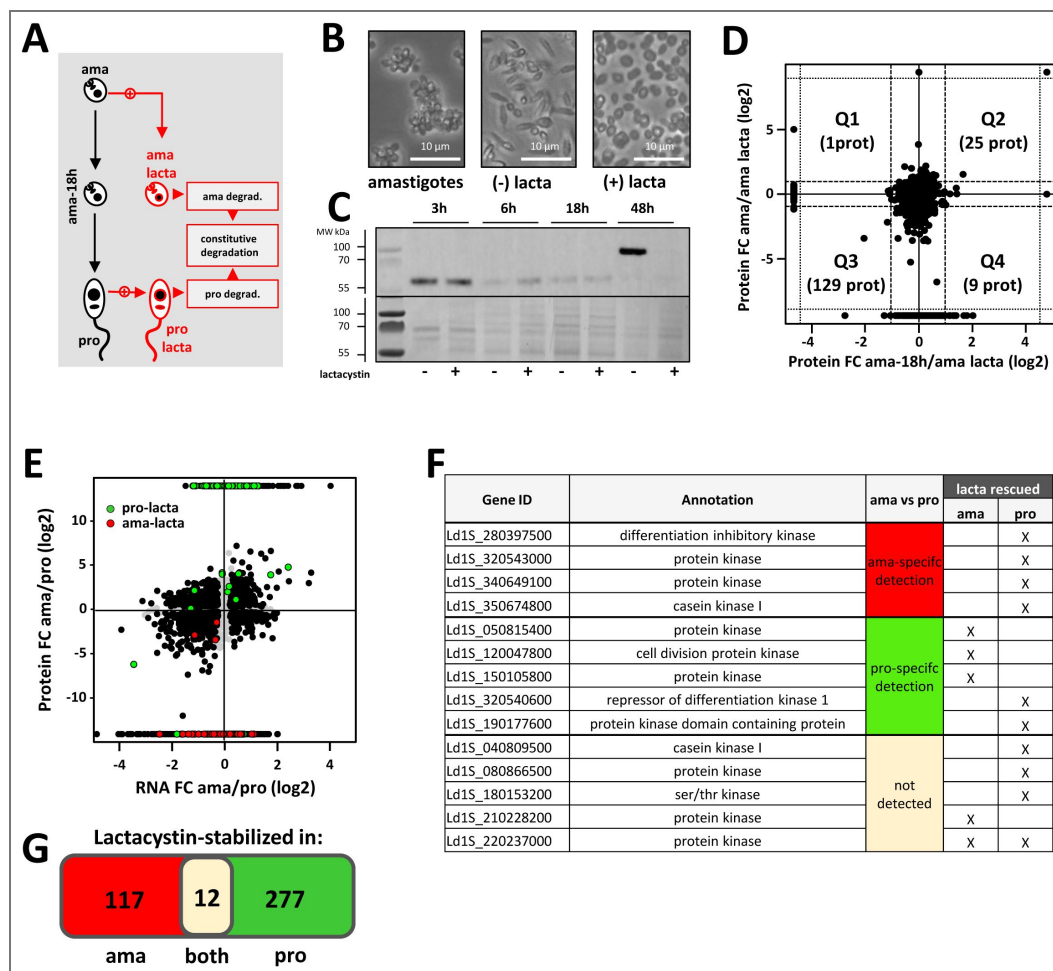


Figure 4. Analysis of stage-specific proteasomal protein turn-over.

(A) Overview of experimental workflow. To assess early and late steps of differentiation (left part of the graph in black), comparative, quantitative proteomics analyses were performed on tissue-isolated amastigotes (ama), amastigotes at 18h into the transition to promastigotes in culture (ama-18h) and fully differentiated promastigotes after 1 passage *in vitro* (pro). To gain insight into stage-specific and constitutive protein degradation (right part of the graph in red), amastigotes and promastigotes were treated with the irreversible inhibitor lactacystin (ama-lacta and pro-lacta). (B) Microscopic images of amastigotes at 18h in absence of lactacystin treatment (amastigotes, left), after 48h in absence (-, middle) or presence of the inhibitor (+, right). (C) Western blot analysis. Protein extracts were obtained from 2×10^6 parasites after 3, 6, 18 and 48h of differentiation from amastigotes to promastigotes in presence (+) or absence (-) of lactacystin. Extracts were separated by electrophoresis, transferred onto PVDF membrane and the presence of PFR2 was revealed using an anti-PFR2 antibody (upper panel). A Coomassie blue stain of the same gel is shown as loading control (lower panel). (D) Double ratio plot comparing the log2 fold changes in protein abundance between ama-18h/ama-lacta (x-axis) and ama/ama-lacta (y-axis). Dashed lines indicate FC = 2 and the infinite symbol (∞) represent proteins that are unique to one or the other comparison. (E) Projection of the proteins stabilized in presence of the inhibitor in ama (ama-lacta, orange) and pro (pro-lacta, pink) onto the double ratio plot shown in Figure 3B. Note that ama-specific proteins are rescued from degradation in pro-lacta, while pro-specific proteins are rescued from degradation in ama-lacta. (F) List of the protein kinases stabilized after lactacystin treatment in pro or ama (columns lacta rescued) and either not detected in the total proteome (column ama vs pro, grey cell) or specifically quantified at the ama (red cell) or the pro (green cell) stage. (G) Venn diagram showing the number of proteins specifically stabilized in ama or pro or stabilized in both stages after lactacystin treatment.

established, stage-specific proteomes (see [Figure 3](#)), an intriguing pattern emerged: the majority of proteins that were rescued in ama-lacta turned out to be part of the promastigote proteome data set ([Figure 4E](#)). *Vice versa*, many proteins rescued in pro-lacta were part of the amastigote-specific proteome data set (Table 9, sheets H and I), a pattern that affected 14 protein kinases ([Figure 4F](#)). Thus, these proteins seem to be expressed constitutively across both stages, while their steady-state stage-specific abundance is regulated by proteasomal degradation. Surprisingly, 12 proteins (including the protein kinase Ld1S_220237000) were rescued from degradation in both ama-lacta and pro-lacta but were never identified in the proteomes of the corresponding untreated controls, suggesting that they are constitutively degraded in both stages under our experimental conditions ([Figure 4G](#), Table 9, sheet F).

In conclusion, our results confirm the important role of protein degradation in regulating the *L. donovani* amastigote and promastigote proteomes and identify protein kinases as key targets of stage-specific proteasomal activities.

Quantitative phosphoproteome analysis reveals *Leishmania* biological networks associated with stage differentiation

The investigations above associated the expression of protein kinases (and their degradation) to *Leishmania* stage differentiation ([Figure 4F](#), S8 and S9), thus confirming previous reports on the importance of phosphotransferase activities in trypanosomatid development ([93–103](#)). Differential turnover of these important signaling proteins by stage-specific proteasomal activities likely represents a regulatory switch controlling parasite development. To further investigate how this putative regulatory switch operates, we screened for stage-specific protein kinase substrates applying label-free, quantitative phospho-proteomic analyses on three independent amastigote and promastigote samples. We identified a total of 2079 phospho-peptides in amastigotes and 7095 phospho-peptides in promastigotes, suggesting a three-fold increase in global protein phosphorylation in insect-stage parasites (Table 10). However, increased TiO₂ phosphopeptide enrichment in a given sample may be due either to increased substrate expression or increased *de novo* phosphorylation ([Figure 5A](#) and [S10](#)). To distinguish between these two possibilities we assessed stage-specific relative phosphorylation changes normalized to protein abundance ([Figure 5B](#)).

We identified 877 phosphoproteins in amastigote samples carrying 2079 phosphosites of which 55.8% (1160) were uniquely identified at this stage, while 12.6% (565) showed a significant change in normalized phosphorylation levels compared to promastigotes ([Figure S10](#), Table 11 sheets A to D). Conversely, 2054 phosphoproteins with 7095 phosphorylation sites were identified in promastigote samples, of which 70% (4967) were exclusively detected at this stage, while 1% ([73](#)) presented a significant change in normalized phosphorylation levels compared to amastigotes ([Figure S10](#), Table 11 sheets F to I). These data reveal a surprising level of stage-specific phosphorylation in promastigotes, which may reflect their increased biosynthetic and proliferative activities compared to amastigotes, or may be a consequence of culture adaptation.

We next investigated the pathways regulated by phosphotransferase activities analyzing the phosphoproteomes of amastigote and promastigote for enriched biological processes ([Figure 5C](#)) (Table 11, sheets E and J). Stage-specific enrichment was observed in the pro dataset for ‘ciliary cell motility’, indicating that flagellar biogenesis and activity are not only regulated by increased protein abundance, but also by increased phosphorylation of flagellar proteins. Surprisingly, in both the ama and pro datasets, similar functions were enriched, including cell division (e.g. ‘cytokinesis’, ‘cell cycle’), signaling (e.g. ‘protein aa phosphorylation’, ‘signal transduction’), or post-transcriptional regulation of gene expression. Thus, aside stage-specific protein turnover (see above, [Figure 4](#)), differential phosphorylation represents yet another layer of regulation that contributes to the establishment of biological networks in *Leishmania* that share similar function in both ama and pro, but whose components are stage-specifically regulated at post-translational levels.

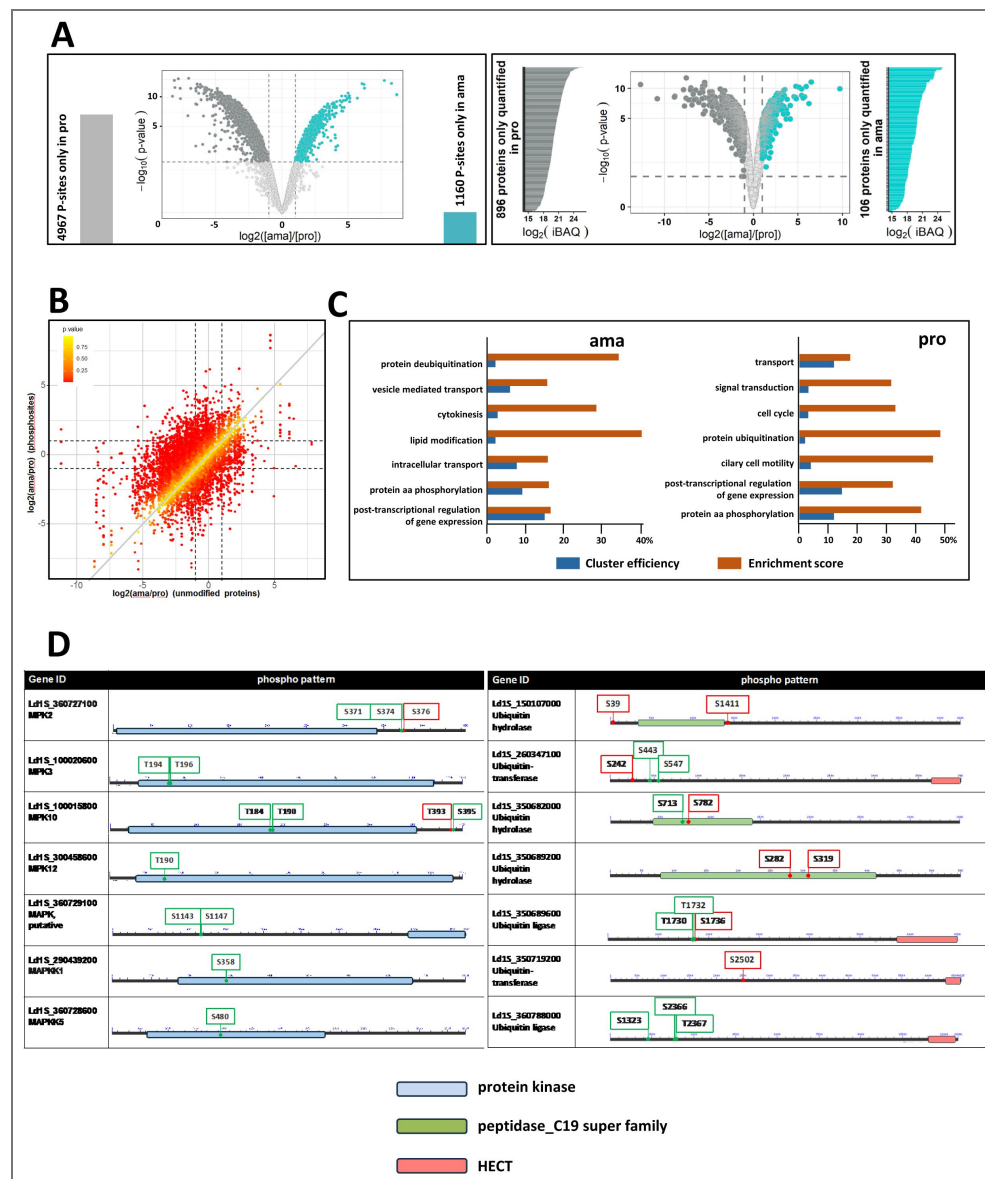


Figure 5. Stage-specific phospho-proteomic profiling.

(A) Volcano plots corresponding to the total proteome (right panel) and phosphoproteome (left panel) analyses. Proteins and phosphosites (P-sites) only identified in one stage are presented at each side of the volcano plots. Cyan dots (ama) and grey dots (pro) represent signals with $FC \geq 2$ and $FDR < 1\%$. Light grey dots represent non-significant changes. (B) Relative phosphorylation change normalized to protein abundance. Ratio plot comparing the $\log_2$ fold changes in total protein abundance between ama and pro (x-axis) and phosphosite abundance in ama versus pro (y-axis). Dashed lines indicate a $FC = 2$. The color intensity of each dot reflects the p-value calculated for the relative phosphorylation change normalized to protein abundance as indicated in the graph. Confidence values were derived as described in Supplementary Methods. (C) GO term enrichment analysis for the category 'biological process'. Only stage-specific phosphosites were considered (i.e. sites that showed a significant increase in relative phosphorylation normalized to protein abundance, and sites that were only detected in one or the other stage, whether the protein was identified or not in the total proteome analysis). The histogram plots show 'cluster efficiency' in blue and 'enrichment score' in orange for the GO term enrichment analysis performed with the ama (middle panel) and pro (right panel) datasets. The main GO terms identified using the BiNGO plug-in in Cytoscape and associated with a BH p-value < 0.05 are shown. (D) Phosphorylation pattern of MAPK and MAPKK proteins (left panel) or Ubiquitin hydrolases and transferases (right panel) identified in the relative phosphorylation change normalized to protein abundance. The positions of the phosphorylated amino acid specific to ama (red) and pro (green) are indicated. Only phosphopeptides identified in the relative phosphorylation change normalized to phosphorylation level (see Table 11) and detected either in one stage or the other or quantified with a $FC \geq 2$ were considered for the identification of the phosphosites.

Analyzing stage-enriched, functional networks in further detail using the STRING applications of the Cytoscape software package revealed a series of potential regulatory interactions (Figure S11 [\[link\]](#), Table 12): A first group of protein kinase substrates is represented by protein kinases themselves with more than 30 and 100 members of this large protein family identified in the ama and pro phosphoproteomes, respectively (Table 11, sheets D and H, see Figure 5D [\[link\]](#) for phosphorylation pattern of some members of the MAP kinase pathway). An additional group of protein kinase substrates in both amastigote and promastigote phosphoproteomes include stage-specific phosphatases (18 in ama and 55 in pro) that can counteract protein kinase activities and even deactivate signaling cascades by dephosphorylation of protein kinase substrates (Figure S11 [\[link\]](#), Table 11, sheets D and H). Furthermore, the stage-specific phosphoproteomes showed enrichment in functional networks linked to ribosomal biogenesis and RNA processing suggesting that protein kinases may control the establishment of stage-specific ribosomes. Finally, the identification of networks linked to proteasomal protein degradation in both ama and pro datasets not only further sustains the role of protein turnover in *Leishmania* stage differentiation, but uncovers a possible feedback loop between proteasomal activities controlling stage-specific degradation of protein kinases (see Figure 4 [\[link\]](#)), which in turn may control stage-specific ubiquitin ligases and deubiquitinating enzymes via differential phosphorylation (see Figure 5D [\[link\]](#) for phosphorylation pattern of selected members of the Ubiquitin proteasomal system).

Discussion

By applying a 5-layer systems analysis to spleen-derived bona fide amastigotes and culture-derived promastigotes, we identified a series of functionally related gene clusters that are co-regulated during *Leishmania donovani* stage differentiation. This co-regulation occurs at multiple levels, including transcript stability, proteasomal protein turnover, and protein phosphorylation. Our study underscores the complex molecular architecture of the *Leishmania* developmental process and provides novel insights into possible regulatory feedback loops that may coordinate stage-specific transitions. These results raise important questions regarding the integration of these regulatory networks during differentiation, and their contributions to environmental sensing and adaptive evolution. In the following we discuss these putative networks in the context of the current literature and propose experimental strategies for their downstream, mechanistic analyses.

A first series of possible regulatory networks proposed by our results could rely on recursive or self-controlling interactions, where components of a given pathway are regulated by the pathway itself (Figure 6A [\[link\]](#)). In promastigotes for example, a co-regulated, functional gene cluster showing stage-specific increase in transcript abundance was defined by the GO term ‘post-transcriptional regulation of gene expression’. In the absence of stage-specific gene dosage changes (Figure 1 [\[link\]](#)) and the lack of transcriptional regulation of gene expression in *Leishmania* (104) the stage-specific expression changes of these transcripts thus is likely regulated itself at the post-transcriptional level. Such auto-regulatory feedback is common for RNA-binding proteins (RBPs), which play a central role in controlling mRNA stability, often targeting their own transcript (105). This helps keep protein levels stable (negative feedback) or create on/off switches for developmental processes (positive feedback) (106). In *T. brucei* for example, the stage-specific expression of RBPs, such as ZC3H22 or RBP9, is regulated at post-transcriptional levels (107), while other RBPs, such as TbZFP3 or RBP10, can control developmental transitions by stabilizing cohorts of mRNAs, thus defining developmental regulons (108, 109). In *Leishmania*, RBPs are likely to participate in comparable regulatory networks. These can be investigated through methods such as crosslinking and immunoprecipitation (CLIP) to map RBP binding to 3' untranslated regions (3'UTRs), computational approaches to identify conserved regulatory sequence motifs, and loss-of-function analyses targeting either trans-acting factors (RBPs and their interaction partners) or cis-acting elements (e.g., RBP binding sites within 3'UTRs).

Our data propose a similar recursive feedback loop that seems to act on the level of protein translation, which was revealed by a paradox: while amastigotes showed increased abundance for 38 transcripts annotated for the GO term ‘ribosome biogenesis’ (Table 4, sheet E), the

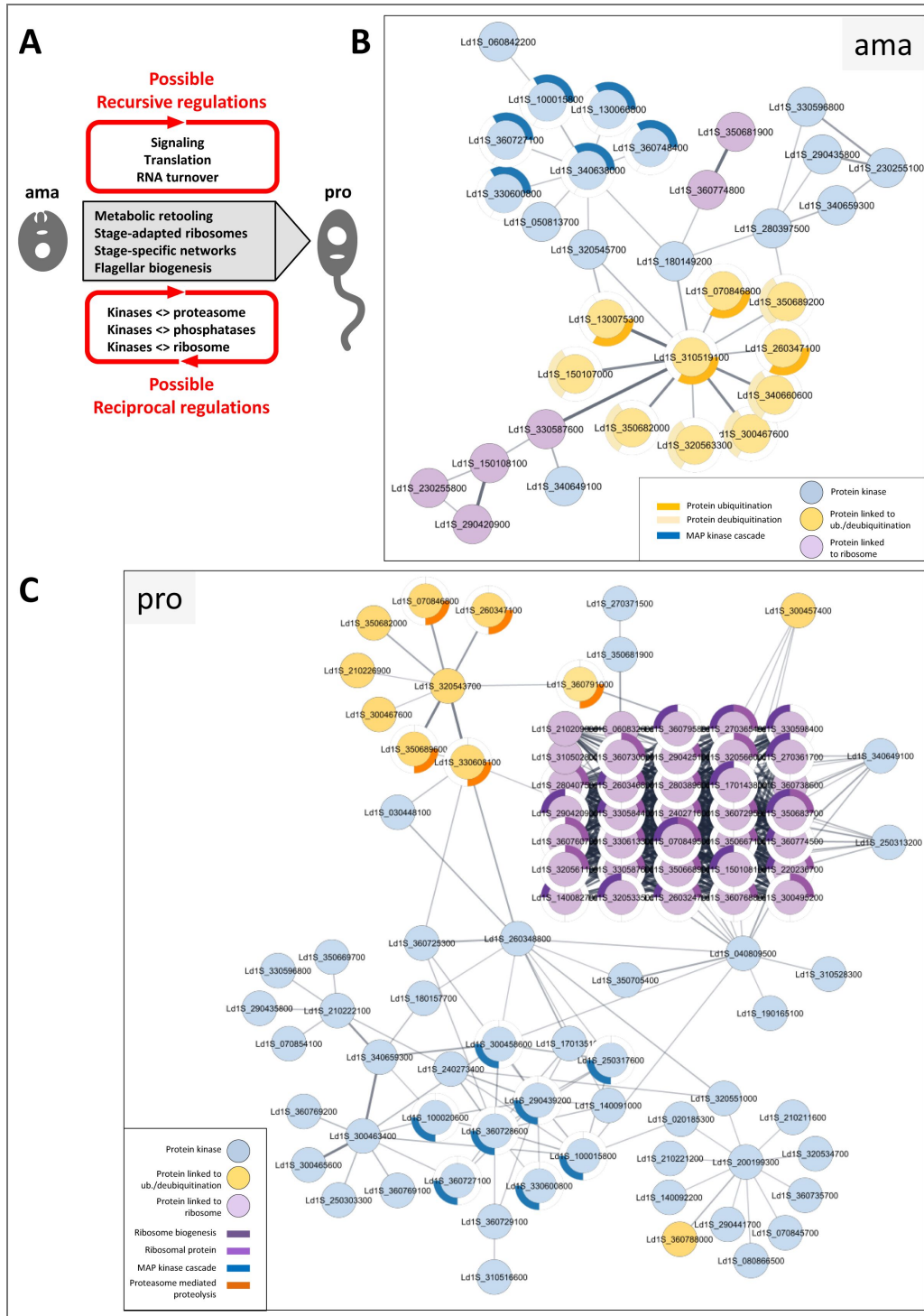


Figure 6. Network analyses.

(A) Model of *Leishmania* gene expression regulation during stage differentiation. (B-C) Network analysis. Networks, restricted to the protein kinases (blue nodes), proteins implicated in ribosome biogenesis and ribosomal proteins (violet nodes) and proteins linked to ubiquitination/deubiquitination (orange nodes) identified based on normalized phosphorylation levels conducted in ama (B) or pro (C), were generated with the STRING plug-in of the Cytoscape software package using *L. infantum* orthologs and full STRING network with a confidence score cutoff of 0.4. Each node represents a phosphoprotein with the respective gene identifier indicated. GO terms associated to the proteins are represented by the colored segments around the nodes according to the legend shown in the graph. Only GO terms associated with a p-value < 0.05 were considered.

corresponding proteins showed reduced abundance in amastigotes compared with promastigotes (Figure 3C). This dissociation between RNA and protein abundances indicates that ribosomal components themselves may be regulated at translational levels. Increased ribosomal protein abundance in promastigotes could satisfy the requirement of increased translation capacity at this fast-growing stage (31, 110). In contrast, increased mRNA abundance of ribosomal components in amastigotes may indicate stage-specific mechanisms of mRNA storage (111) that may allow to jump start promastigote differentiation and accelerated growth following exposure to low temperature and neutral pH. This possibility is supported by our proteomics investigations of the early differentiation process, revealing ribosomal biogenesis as one of the first pathways induced over the initial 18h during the amastigote-to-promastigote developmental transition (Figure S6).

Differentiation to promastigotes was associated not only with quantitative changes of ribosomal components as indicated by co-regulated expression of ribosomal proteins and translation factors (see Table 8), but also seems to affect ribosome structure itself as indicated by stage-specific expression changes of gene sets enriched in the GO terms ‘RNA methylation’ (8 methyltransferases, including the rRNA methyltransferases Ld1S_350672300 and Ld1S_350694000) and ‘pseudouridine synthesis’ (6 pseudouridine synthases and the H/ACA snoRNP Nop10 (Ld1S_360722900) (Table 8, sheets D to H). The differential expression of snoRNAs (see Figure 2C and Table 5) and the developmentally regulated changes in rRNA modification that can affect ribosome activity (34) (Figure 2D) indeed support the existence of such stage-specific ribosomes. These results are in line with our recent reports providing indication for specialized ribosomes in *Leishmania* adaptation, in which we correlated changes in rRNA pseudouridylation to *L. donovani* fitness gain in culture (31, 66). Cryo-EM analysis of purified ribosomes from amastigotes and derived promastigotes, combined with targeted deletion of individual snoRNA genes, will be critical to uncover stage-specific structural and functional adaptations of such stage-specific ribosomes.

Finally, a third possible recursive feedback loop seems to act at the post-translational level involving members of various protein kinase (PK) families, which not only showed stage-specific expression, but also represented a main class of phosphorylation substrates themselves. This interaction may reflect stage-specific signaling cascades, where downstream kinases are activated through phosphorylation by upstream kinases, such as exemplified by the MAP kinase pathway (103, 112). Indeed, 11 members of the MAP kinase family showed stage-specific changes in their phosphorylation pattern (Figure 5D), 9 of which were only detected in the promastigote phospho-proteome and were previously associated with this stage, being implicated in flagellar biogenesis (MPK9 and MPK3) or metacyclogenesis (MPK4) (98, 102, 113). In contrast, the two MAP kinases MPK2 and MPK10 - previously linked to amastigote differentiation and virulence (99, 102) - showed an amastigote-specific increase in protein abundance and relative phosphorylation change for defined phosphorylation sites. In *L. mexicana*, such a recursive PK feedback loop has been identified between the interacting protein kinases LmxMKK and LmxMPK3, which cross-phosphorylate each other to regulate flagellar length (113). To further dissociate the complex regulatory relationships between parasite PKs, protein-protein interaction maps should be established (e.g. co-IP, proximity labelling), combined with in vitro kinase assays to reveal direct kinase-substrate relationships.

Most PKs have pleiotropic effects across biological systems, due to their ability to phosphorylate a wide range of functionally unrelated substrates (114). Our analysis of relative phosphorylation changes normalized to protein abundance confirms such pleiotropic function in *Leishmania* and suggests parasite PKs at the center of a second form of possible feedback loop defined by reciprocal interactions, where components of two or more regulatory pathways may cross-control each other (Figure 6A). For example, as judged by changes in phosphorylation abundance, it seems that stage-specific PK activities act on a series of phosphatases, which in return may modify kinase activities by dephosphorylation (Table 11, sheets D and G). Such reciprocal regulation between both enzyme families is well-established across many eukaryotic systems, for example between mammalian ERK1/2 and the MAPK phosphatase DUSP6/MKP-3 (115), or yeast CDK1 and the phosphatase Cdc25, controlling mitotic entry (116). In addition, 47 differentially phosphorylated proteins were associated with ‘Translation’, some of which have been linked in

Leishmania to translational control, cell cycle regulation or infectivity (29, 117–121). This opens the possibility of yet another reciprocal feedback loop, where phosphorylation of ribosomal components could fine tune translational control, which in turn may affect the expression of protein kinases themselves (122–130).

Finally, our results suggest the possibility of a third reciprocal feedback loop between protein kinases and proteins of the proteasomal system that may govern *Leishmania* stage transitions. Previous genetic analyses demonstrated an essential role for various components of this system in promastigote-to-amastigote transition and intracellular survival, including *L. mexicana* ubiquitin conjugating (E2) enzymes (UBC1/CDC34, UBC2 and UEV1), the E3 ubiquitin ligase HECT2 or various deubiquitinases (e.g. DUBs 4, 7, 13) (91, 92). By applying the irreversible proteasome inhibitor lactacystin, we extended this role to the reverse process - amastigote-to-promastigote differentiation - demonstrating that proteasomal activities are critical for both directions of the developmental cycle (Figure 4C). The constitutive expression of many proteasomal components across both amastigote and promastigote stages supports a role of phosphorylation in regulating stage-specific proteasomal activities, considering that phosphorylation of DUBs has been linked to changes in their stability, localization, specificity or catabolic activity (131, 132). We indeed identified 11 DUBs and 10 ubiquitin ligases/transferases showing stage-specific phosphorylation at specific residues (Table 11, sheets D and I). This correlated with the stage-specific proteasomal degradation of 14 PKs (Figure 4F, Table 9, sheet G), including two previously implicated in *Leishmania* *in vivo* fitness (Repressor of differentiation kinase 1, Ld1S_320540600; differentiation inhibitory kinase, Ld1S_280397500) (102). These findings support the existence of a possible kinase/proteasome reciprocal feedback loop in early parasite development. The experimental validation of such regulatory interactions will require in-depth biochemical, molecular, and systems biology approaches involving global kinase-substrate mapping (114), *in vitro* reconstitution assays using recombinant proteins to demonstrate reciprocal kinase and proteolytic activities, and genetic perturbation test to reveal the biological impact of the regulatory interaction.

The complexity of regulatory interactions between protein kinases and the proteasomal pathway is further increased by observations showing that substrate phosphorylation can mediate subsequent ubiquitination and degradation, while ubiquitination of MAP kinase family members can regulate kinase activity and localization rather than promote degradation (139–141). Future experiments simultaneously analyzing changes in both phosphoproteome and ubiquitome during *Leishmania* stage-differentiation and making precision mutants that lack the ubiquitination or phosphorylation sites may help elucidate this complex regulatory interaction and identify individual proteins and their PTMs that drive parasite development

In conclusion, our data represent an important novel resource to experimentally assess the regulatory landscape of *Leishmania* stage development. We propose genetic feedback control as a central mechanism in parasite differentiation, with stage-specific interactions between genes and their products likely adapting the parasite phenotype to the environmental changes encountered inside its vertebrate and invertebrate hosts. In contrast to the major phenotypic shift observed between mammalian-stage amastigotes and insect-stage promastigotes, each developmental stage is stably maintained within its respective host even in the wake of environmental fluctuations. Such phenotypic robustness is known to rely on highly inter-connected, stage-specific regulatory networks (142), which in the case of *Leishmania* could implicate recursive and reciprocal genetic interactions between protein translation, protein degradation and protein phosphorylation (Figure 6B and C). Our results define *Leishmania* differentiation as an excellent model system to study how alternations between phenotypic plasticity (i.e. developmental transitions) and robustness (stable maintenance of stages) shape microbial fitness, and provides a powerful experimental framework for future studies on the emergence of stable, adaptive biological traits through feedback regulation.

Sup figure legends

Hamster	Analysis				
	DNAseq	RNAseq	Proteomic+/-lacta	Metabolomic	Proteomic/ Phosphoproteomic
H-1	ama1/pro1				
H-2	ama2/pro2				
H-4	ama3/pro3				
ERC9-1		ama4/pro4			
ERC9-2		ama5/pro5			
ERC4-2		ama6/pro6			
H313-1		ama7/pro7			
H208-2			ama8/pro8		
H211-1			ama9/pro9		
H211-2			ama10a*/pro10a** & ama10b*/pro10b**		
H131-1				ama11/pro11a** & pro11b**	
H131-2				ama 12/pro12a** & pro12b**	
H133-1				ama13	
H133-2				ama14	
H130-1					ama15
H130-2					ama16
H134-1					ama17
H134-2					pro13a** & pro13b**
H138-2					pro14
H139-2					ama18/pro15

Foot note: * amastigote preparation from the spleen of one infected hamster was divided in 2, ** promastigote cultures derived from the same initial pool of splenic amastigotes that was split in 2 flasks for differentiation and culture until passage 2.

Figure S1. Overview of the samples used in this study.

Each *L. donovani* infected hamster is identified by the cage number, as are the hamster-derived amastigotes (ama) and corresponding promastigotes (pro).

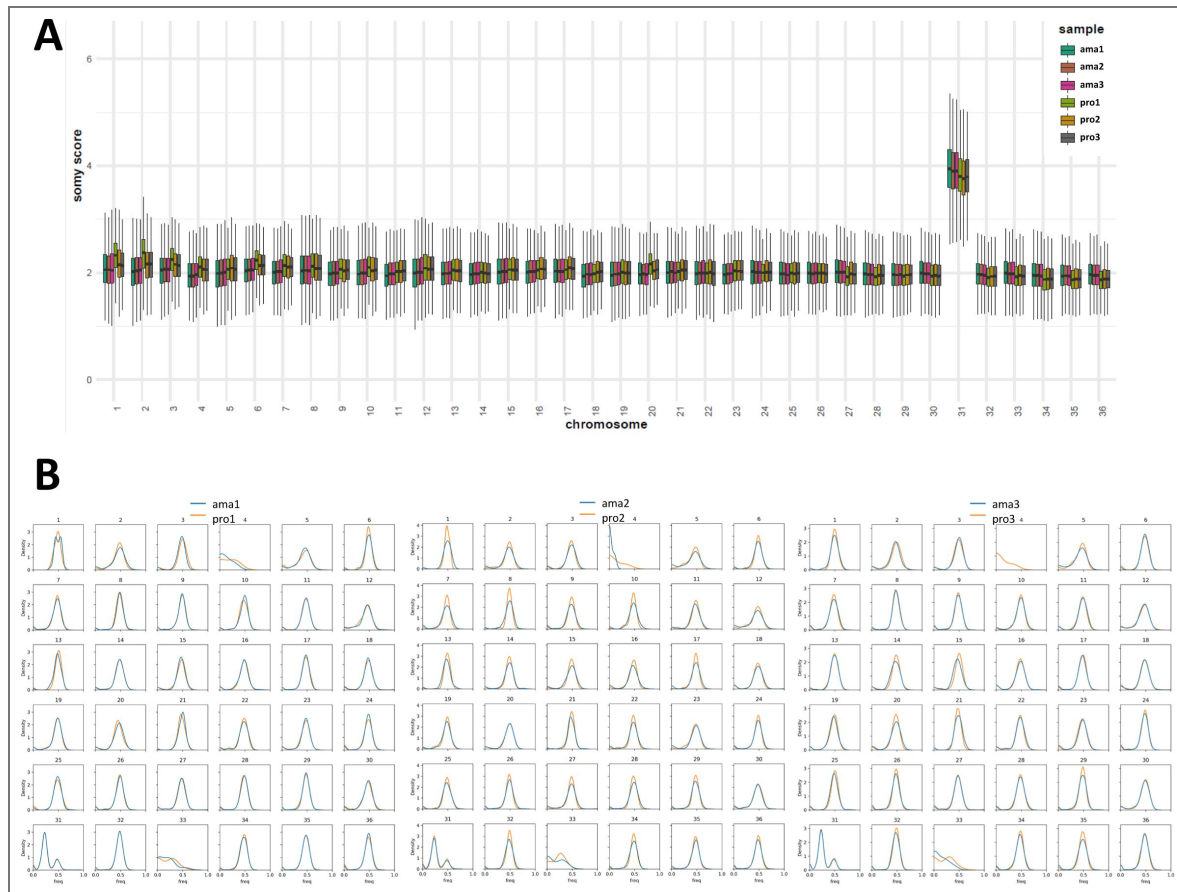


Figure S2.

(A) Box plots representing the median somy score for all the chromosomes. Each box corresponds to one of three biological replicates for ama and pro as indicated in the graph. Sample identifiers are detailed in the legend of Figure 1A and Figure S1. (B) Frequency distribution plots representing, for each biological replicate and each chromosome, a kernel density estimate of the distribution of SNP frequencies in the amastigote and promastigote sample.

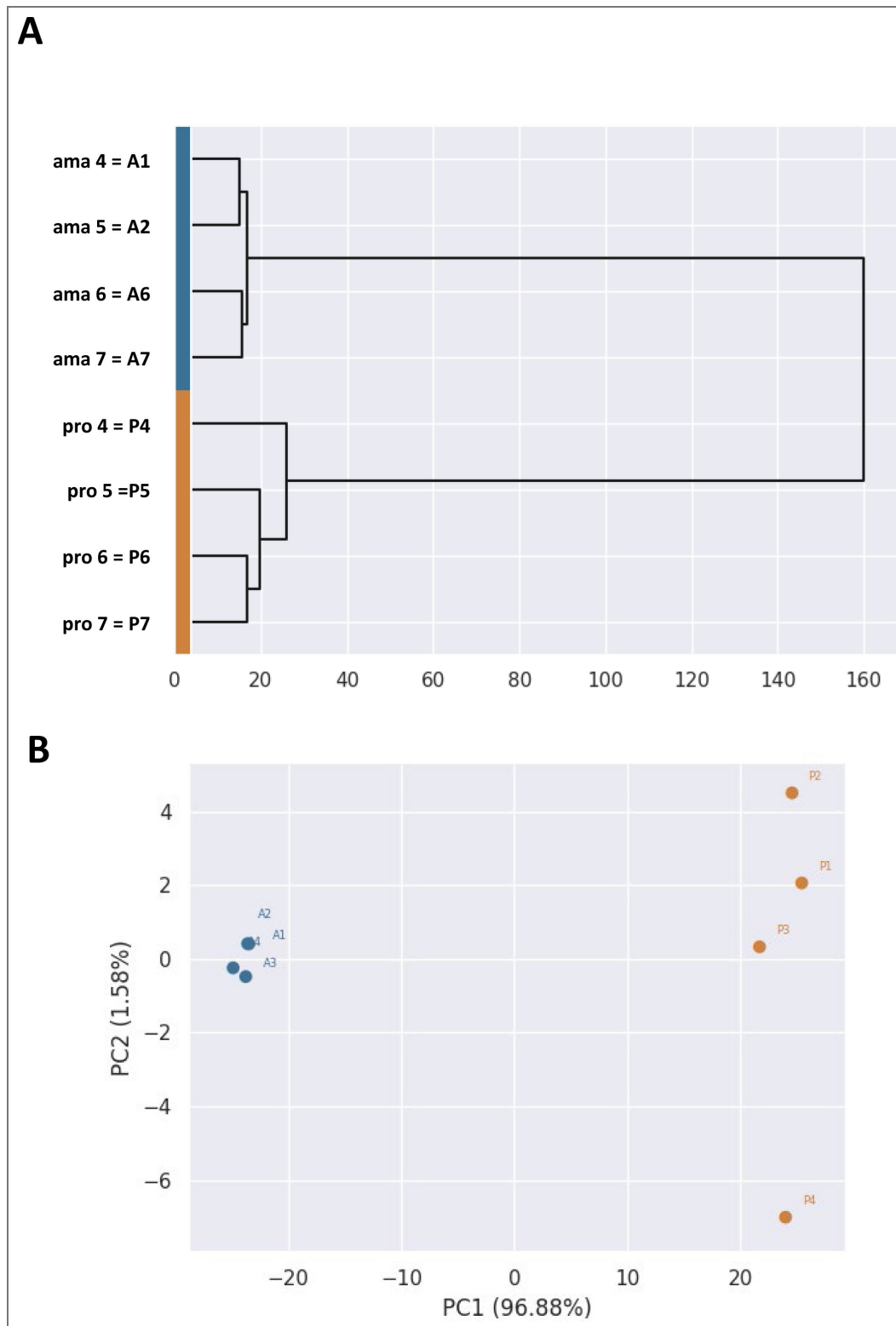


Figure S3.

(A) Hierarchical clustering of the samples. The dendrogram is built using the Ward method. The normed data was log-transformed first and at max 5,000 most variable features were selected. (B) Principal component analysis. The two main components are represented. Ama and pro samples correspond to blue and orange dots respectively.

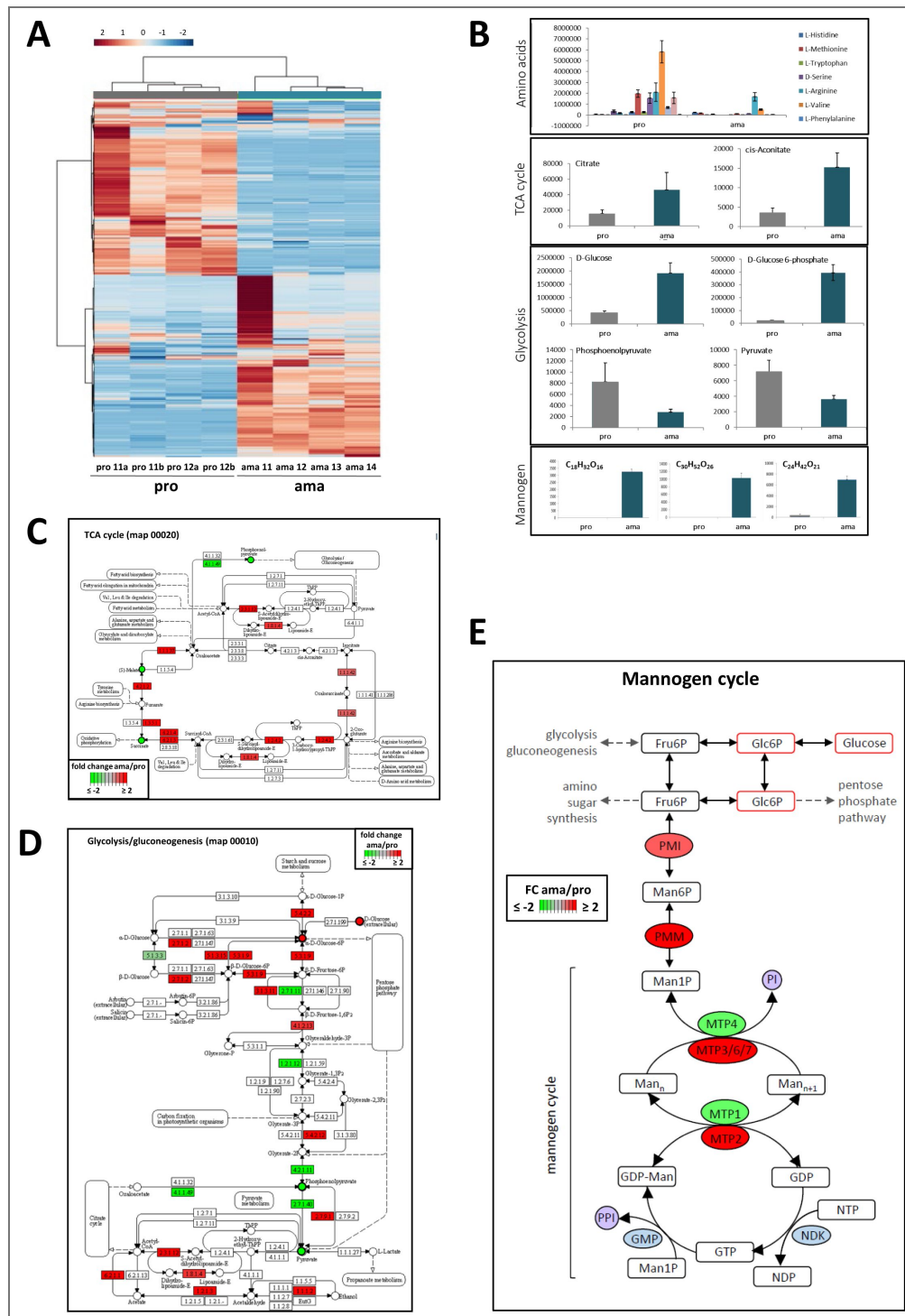


Figure S4.

(A) Cluster analysis of metabolites showing differential abundance in ama and pro samples. (B) Quantification of amino-acids or metabolites involved in glycolysis, TCA cycle and mannogen cycle. The y-axis represents the intensity for each metabolite as detected by MS. (C-D) Correlation between metabolomic and proteomic analyses for TCA cycle (C) and Glycolysis/Gluconeogenesis (D). Fold changes in proteins abundance (squares) and metabolite abundance (circles) with adjusted p-values of respectively < 0.01 and < 0.05 were projected on the respective KEGG maps. The color intensity reflects the fold change value as indicated by the legends in the graphs. (E) Representation of the mannogen pathway. Metabolites are represented by a square, with red-lined squares corresponding to the metabolites that accumulate at the ama stage. Proteins are represented by a circle, with fold changes in protein abundance between ama and pro datasets indicated by color and intensity (red, increase; green, decrease).

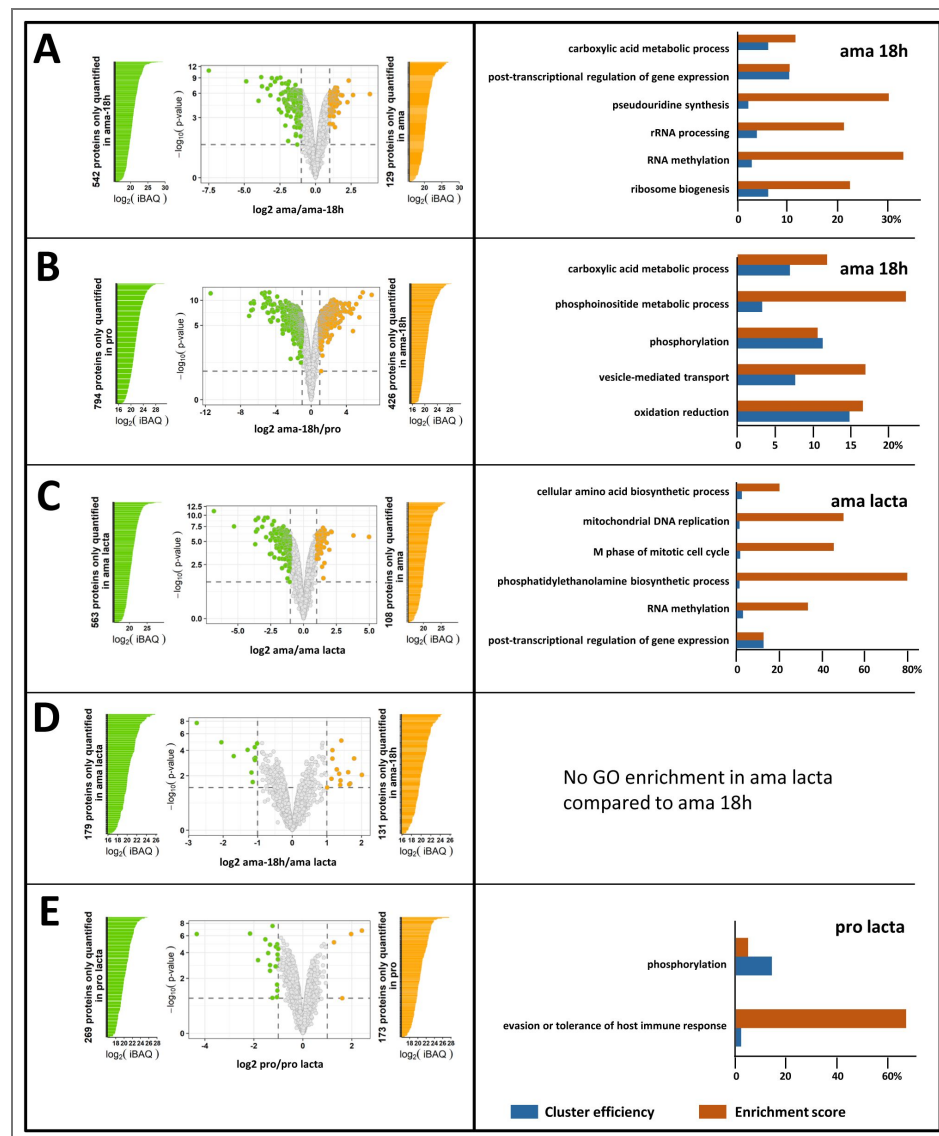


Figure S5. Label free quantitative proteomic analyses in presence or absence of lactacystin.

(A) to (E) Volcano plots representing changes in protein abundance for the different comparisons (left panels) and GO term enrichment analyses for the indicated sample (right panels). Proteins identified by at least two peptides in at least three out of four biological replicates were considered. For volcano plots, colored dots indicate values with $FDR < 0.01$ and $FC \geq 2$ (see Table 8). The light grey dots indicate non-significant expression changes. The colored bars indicate unique protein identifications in one (green) or the other (orange) condition with relative abundance indicated by the iBAQ value (left panels). Proteins quantified in only one condition or showing a significant increase in abundance ($FC \geq 2$ and adjusted $p\text{-value} < 0.01$) were used to perform the GO analysis for the category 'biological process'. The histogram plots show 'cluster efficiency' in blue and 'enrichment score' in orange (right panels) for a selection of GO terms identified using the BiNGO plug-in in Cytoscape and associated with a BH $p\text{-value} < 0.05$. (A) ama vs ama-18h; (B) ama-18h vs pro; (C) ama vs ama lacta; (D) ama-18h vs ama lacta; (E) pro vs pro lacta.

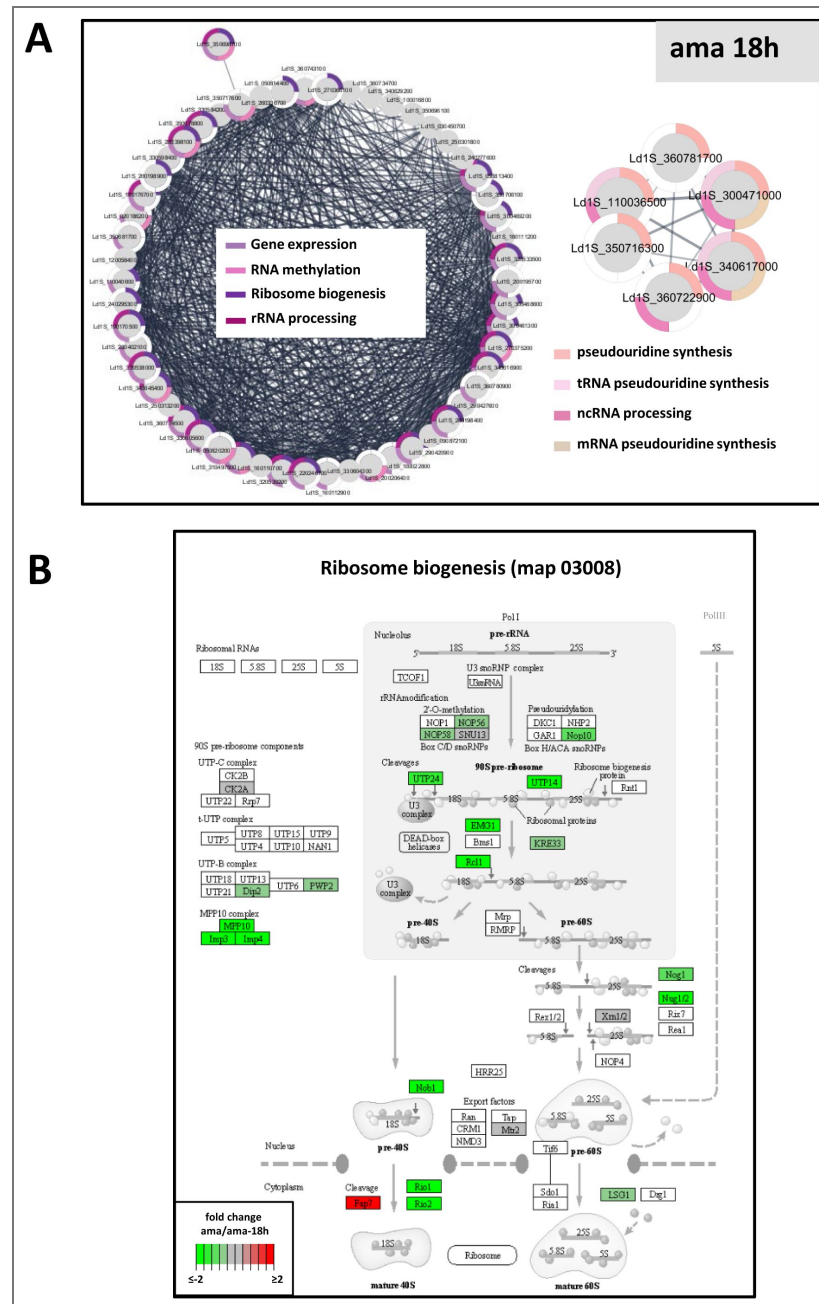


Figure S6.

(A) Functional network enrichment analysis for proteins showing increased abundance in ama-18h compared to ama. The network was generated with the *L. infantum* orthologs using the STRING plug-in of the Cytoscape software package and considering the full STRING network with a confidence score cutoff of 0.4. The GO terms associated with each protein are represented by the colored segments according to the legend. Only GO terms associated with a p-value < 0.05 were considered. (B) KEGG map for ribosome biogenesis showing changes in abundance protein between ama and ama-18h. Only proteins with an adjusted p-value < 0.01 were considered. Differential abundance is indicated by the color and its intensity (red, increase; green, decrease).

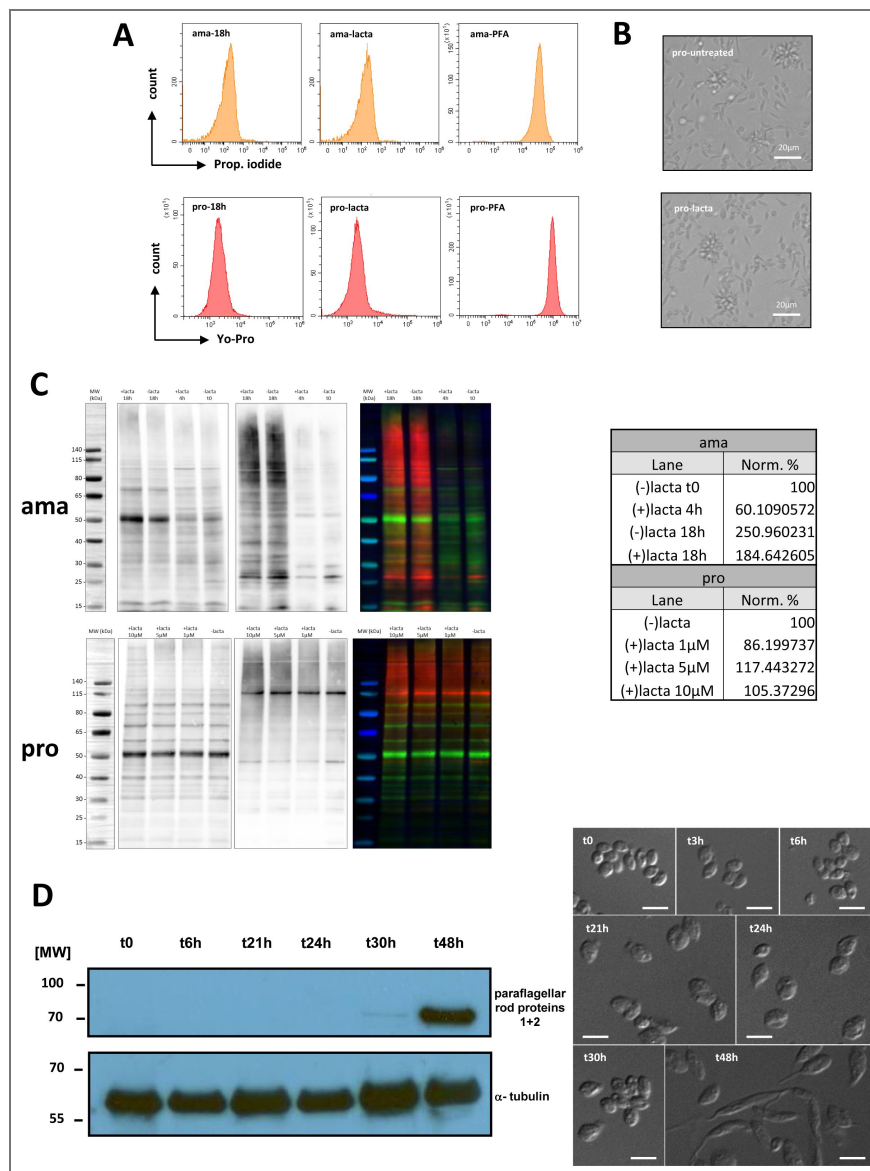


Figure S7.

(A) Viability assessment of the indicated amastigote (upper panel) and promastigote (lower panel) samples after lactacystin treatment by FACS analysis using propidium iodide or YoPro staining. (B) Microscopic images of promastigotes at 18h in absence (upper image) or presence of 10 μ M of lactacystin (lower image). (C) Western blot analysis of protein extracts obtained from ama (upper image) and pro (lower image) in presence or absence of lactacystin treatment. 10 μ g of proteins were labelled with Cy5, separated on NuPAGE 4-12%, and transferred onto PVDF membrane. Membranes were subsequently incubated with a monoclonal antibody against ubiquitin and revealed with HRP-conjugated secondary antibodies. Images of Cy5-labelled proteins (left images), ubiquitinated proteins (middle images) and an overlay of both images (right images, Cy5-labelled proteins in green, ubiquitinated proteins in red) are presented. The table shows the relative quantification after normalization using the Cy5-labelled proteins of untreated samples as a loading control. (D) Western blot analysis of protein extracts obtained at the indicated time points after *in vitro* differentiation from hamster-derived amastigotes to promastigotes. 10 μ g of protein were separated on NuPAGE 4-12%, transferred onto PVDF membrane subsequently incubated with antibodies against paraflagellar rod proteins 1 and 2 and alpha-tubulin, and revealed with HRP-conjugated secondary antibodies (left image). Micrographs showing hamster-derived amastigotes during *in vitro* differentiation to promastigotes (right image). Parasites were collected at the time points indicated, fixed in 4% PFA, and seeded on poly-L-lysine-treated coverslips. Parasites were observed with a 63x oil immersion objective and images were processed with AxioVision Rel.4.8 software. Differential interference contrast (DIC) microscopy was performed with AxioPlan 2 imaging microscope using a 63x oil immersion objective, Axiovision software and AxioCam MRm camera (Carl Zeiss). The scale bars correspond to 1 μ m.

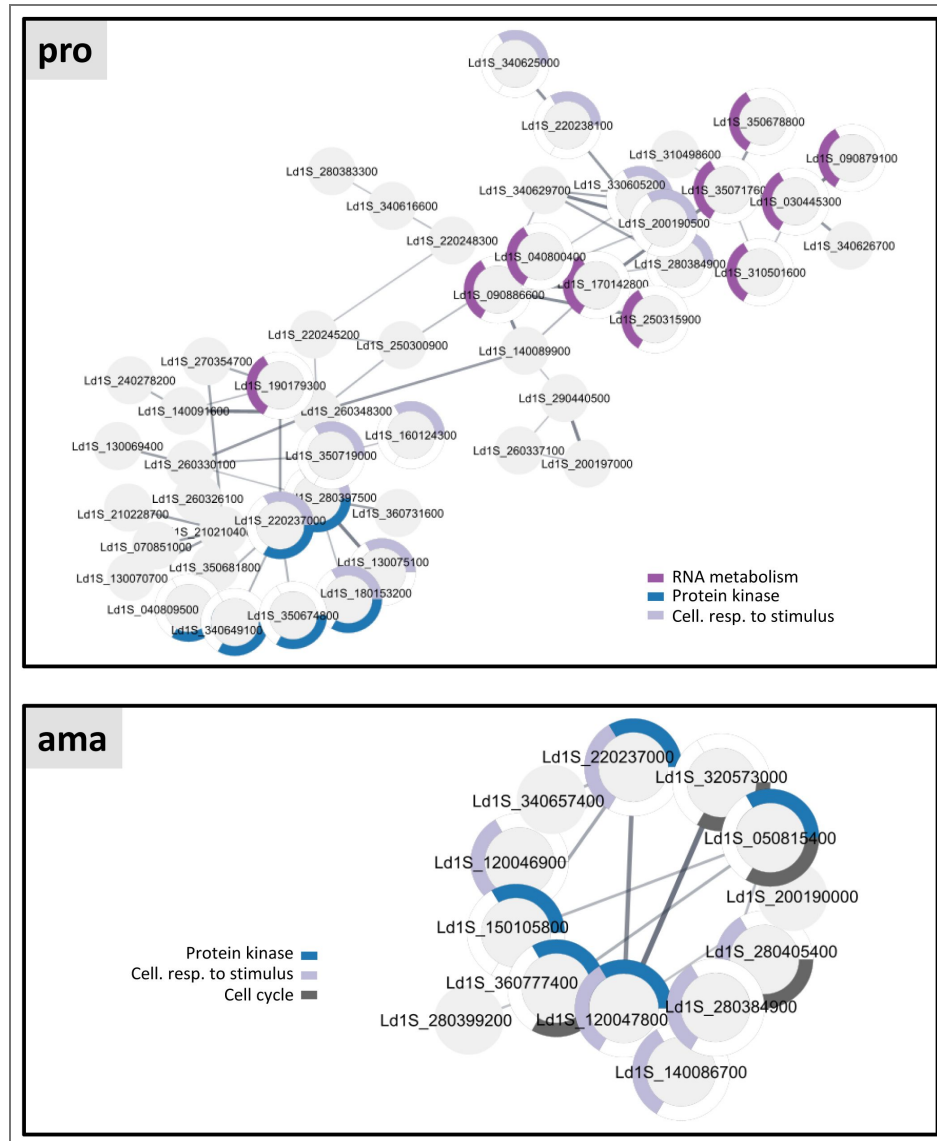


Figure S8. Functional network enrichment analysis for the proteins stabilized in presence of lactacystin in promastigote (pro) and amastigote (ama) datasets.

The network was generated with the *L. infantum* orthologs using the STRING plug-in of the Cytoscape software package and considering the full STRING network with a confidence score cutoff of 0.4. The GO terms associated with each protein are represented by the coloured segments according to the legend. Only GO terms associated with a p-value < 0.05 were considered.

ID Ld1S	ID LdBPK	ID LmxM	Annotation	KO phenotype*
Ld1S_320540600	LdBPK_320860.1	LmxM.31.0810.1	Repressor of differentiation kinase 1, RDK1	Loss of fitness during pro to ama differentiation Loss of fitness in animal infection
Ld1S_220244300	LdBPK_220970.1	LmxM.22.1150.1	Protein kinase domain containing protein, putative	
Ld1S_190177600	LdBPK_191290.1	LmxM.19.1300	hypothetical protein, unknown function (PP: Protein kinase domain containing protein, putative)	
Ld1S_180153200	LdBPK_180650.1	LmxM.07.0900	serine/threonine kinase-like protein, putative	
Ld1S_080866500	LdBPK_080870.1	LmxM.08.0930.1	protein kinase, putative	
Ld1S_040809500	LdBPK_041230.1	LmxM.04.1210.1	casein kinase I, putative	
Ld1S_350674800	LdBPK_351020.1	LmxM.34.1000.1	casein kinase I, putative	
Ld1S_340649100	LdBPK_342850.1	LmxM.33.3020.1	protein kinase, putative	Loss of fitness in animal infection
Ld1S_320543000	LdBPK_321080.1	LmxM.31.1020.1	protein kinase, putative	
Ld1S_280397500	LdBPK_281800.1	LmxM.28.1670.1	differentiation inhibitory kinase, putative, ZFK	Moderate loss of fitness in animal infection
Ld1S_220237000	LdBPK_220370.1	LmxM.22.0490.1	protein kinase, putative	ESSENTIAL
Ld1S_210228200	LdBPK_210330.1	LmxM.21.0270.1	protein kinase, putative	Loss of fitness in animal infection
Ld1S_150105800	LdBPK_151180.1	LmxM.15.1200	protein kinase, putative	
Ld1S_120047800	LdBPK_120130.1	LmxM.12.0130.1	Cell division protein kinase, putative	
Ld1S_050815400	LdBPK_050550.1	LmxM.05.0550.1	protein kinase, putative	

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Figure S9. List of protein kinases stabilized in presence of lactacystin.

Their corresponding orthologs in *L. donovani* strain LdBPK and *L. mexicana* are indicated, as well as the knock-out phenotype as published by Baker et al (2021) (102).

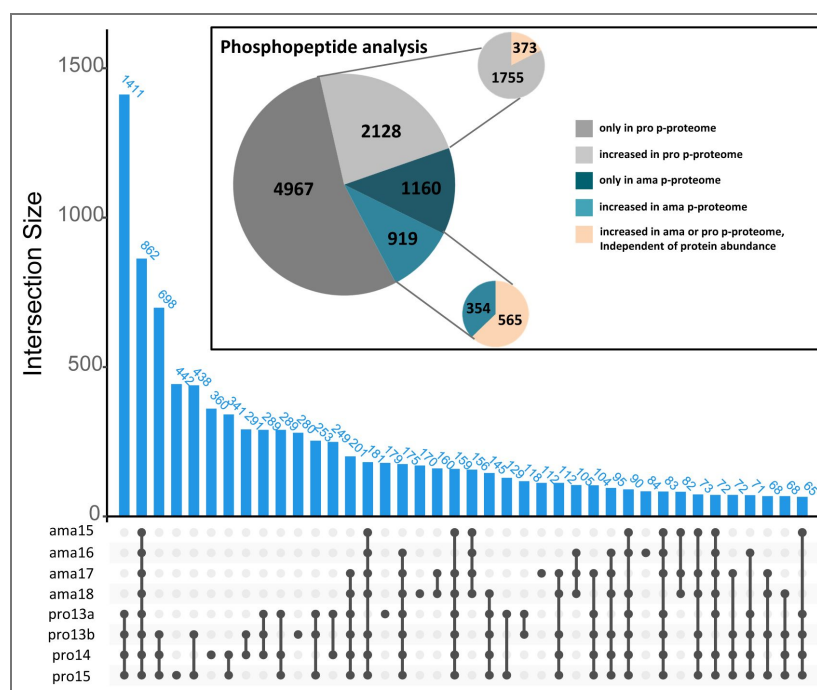


Figure S10. Upset plot representing the number of phosphopeptides (y-axis, blue histograms, numbers are indicated) and the samples in which they were quantified (x-axis).

Pie chart of the phosphopeptide analysis showing the distribution of the phosphorylation sites according to the condition in which they were quantified as indicated in the legend of the graph.

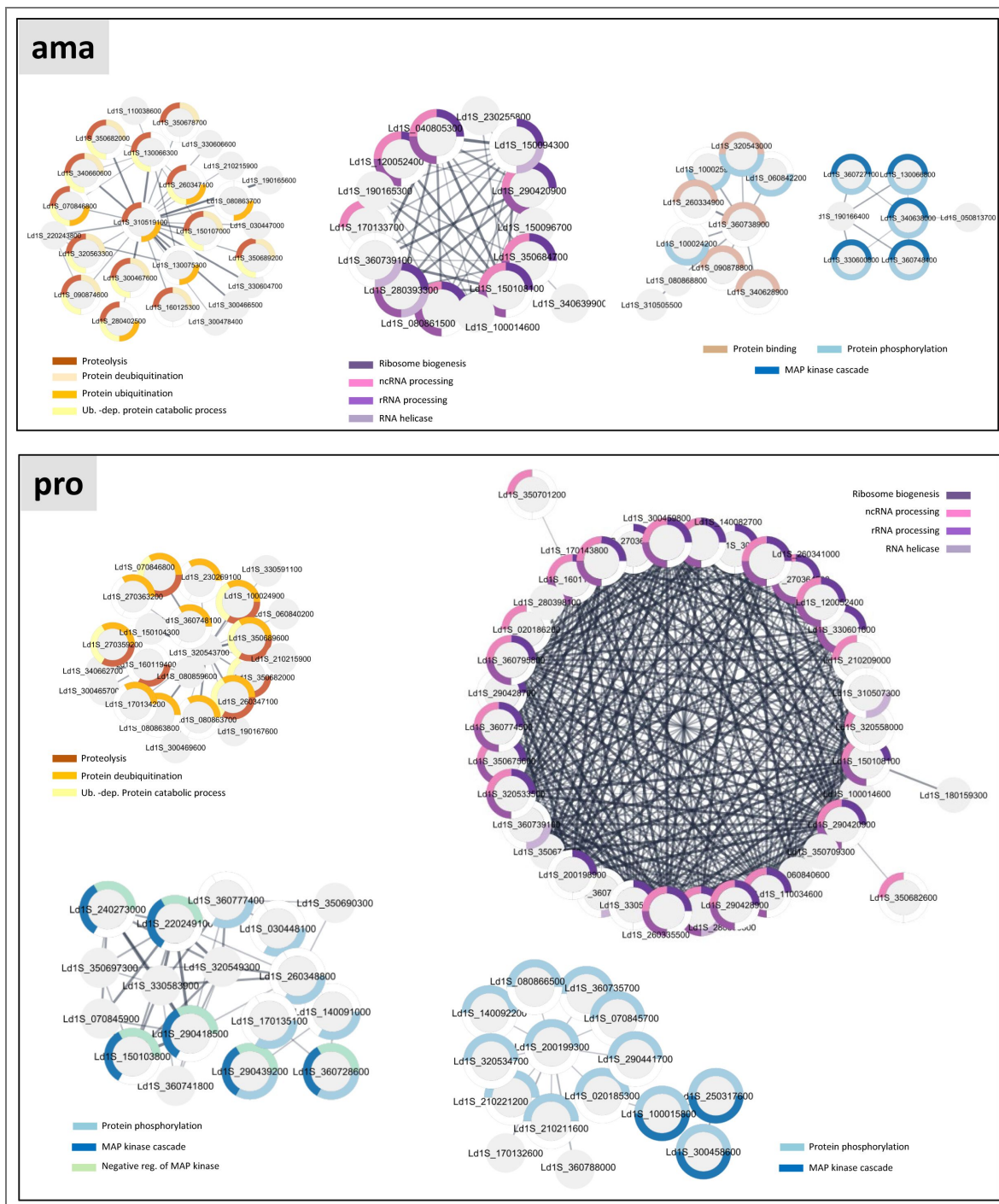


Figure S11. Functional Network analysis.

Networks for proteins that show unique or increased phosphorylation in ama (upper panel) and in pro (lower panel) datasets were generated with the STRING plug-in of the Cytoscape software package using *L. infantum* orthologs and considering the full STRING network with a confidence score cutoff of 0.4. The nodes represent phosphoproteins associated with one or more GO terms (listed in the graph), with the respective gene identifier indicated. The GO terms associated to the proteins are represented by the colored segments around the nodes according to the legend shown in the graph. Only GO terms associated with a FDR < 0.05 were considered.

Data availability

The mass spectrometry proteomics data were deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifiers PXD035697 (phosphoproteome data) and PXD035698 (proteome upon lactacystin treatment). The DNaseq data have been deposited in NCBI's Gene Expression Omnibus and are accessible in the BioProject PRJNA1231373 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1231373>). The RNAseq data have been deposited in ArrayExpress (EMBL-EBI) (69) under the accession number E-MTAB-16528. The mass spectrometry metabolomic data are available on Figshare at <https://doi.org/10.6084/m9.figshare.31027849>. The scripts for GIP and giptools are available at https://github.com/susefranssen/Global_genome_diversity_Ldonovani_complex.

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Additional files

Supplementary data. [↗](#)

Table 1. [↗](#)

Table 2. [↗](#)

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Table 8. [↗](#)

Table 8b. [↗](#)

Table 9. [↗](#)

Table 10. [↗](#)

Table 11. [↗](#)

Table 12. [↗](#)

Table 13. [↗](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors describe co-regulated gene modules underlying stage differentiation in *Leishmania donovani* through a system-level analysis of multiple molecular layers. Using amastigotes isolated from infected hamster spleens and corresponding culture-derived promastigotes, they analyzed genomic variation, transcript abundance, protein levels, phosphorylation states, and metabolite profiles. By combining these, the study identified potential regulatory mechanisms associated with parasite differentiation and generated hypotheses regarding how gene expression is coordinated across different levels.

Strengths:

A major strength of the study is the breadth of the dataset generated. The integration provides an unusually comprehensive view of molecular changes associated with *Leishmania* differentiation in vitro. Such multi-layer datasets involving bona fide vertebrate host stages remain relatively rare in parasitology and will likely become a valuable resource for the molecular parasitology community. In addition, the use of amastigotes isolated from

infected hamsters rather than relying on axenic models provided a biologically relevant framework for the analyses.

The revised manuscript improved several aspects of the original. The RNA-seq analysis is described with a clearer pipeline, and several claims regarding causal regulatory feedback associations have been appropriately toned down. Among the observations reported, the association between parasite differentiation and proteasome-mediated protein degradation is particularly remarkable. The combination of quantitative proteomics with pharmacological inhibition of the proteasome with lactacystin provides support for a role for protein turnover in developmental transitions and paves the way for future mechanistic studies.

Weaknesses:

Most regulatory interpretations remain largely inferential or indirect. The integration identifies correlations between different levels, but direct functional validation is limited/absent. Many of the descriptions should not be interpreted as validated. As highlighted by the authors in this revised version, the mechanistic studies will be part of future work and are beyond the scope of the current work. Of note, the attempt to confirm lactacystin-induced inhibition of proteasomal activity via anti-polyUb immunoblotting did not demonstrate the expected outcome of increase in overall poly-ubiquitylation.

Comments on revised version:

The authors have appropriately addressed my comments and questions from the initial review process. My remaining concern relates to the lack of evidence to confirm proteasomal inhibition by lactacystin in both promastigotes and amastigotes. The immunoblotting experiment newly presented does not reveal a clear increase in the levels of poly-ubiquitylated proteins in treated parasites. In fact, poly-Ub levels were lower at both the 4h and 18h timepoints of treatment. If alternative antibodies or additional immunoblots are not available, the manuscript would benefit from an expanded discussion of this observation and potential explanations. In particular, the interpretation that lactacystin stabilizes ama- and pro-specific degradation would be greatly strengthened by such validation.

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Reviewer #2 (Public review):

Pescher and colleagues present a revised manuscript detailing the multi-omic characterisation of *Leishmania donovani* amastigote to promastigote differentiation and integration of this data. The molecular pathways that regulate *Leishmania* life-stage transitions are still poorly understood, with many approaches exploring single proteins/RNAs etc in a reductionist manner. This paper takes a systems-scale approach and does a good job of integrating the disparate -omics datasets to generate hypotheses about the intersections of regulatory proteins that are associated with life-cycle progression. The differentiation step studied is from amastigote to promastigote using hamster-derived amastigotes which is a major strength. The use of hamsters permits the extraction of parasites that are host adapted and represent "normal", host-adapted *Leishmania* ploidy; the promastigote experiments are performed at a low passage number. Therefore, this is a strength of the work as it reduces the interference from the biological plasticity of *Leishmania* when it is cultured outside the host for prolonged periods. The multi-omics datasets presented are robust in their acquisition and analysis and will form an excellent resource for researchers studying the molecular events (particularly proteasomal protein degradation, and phosphorylation) during life-stage progression.

General comments on the revisions:

My view is that the authors have made significant, satisfactory changes that address the comments and queries I made on the original manuscript (Review Commons).

There are two areas where the authors had to make major changes/justifications where further comment is merited, these were:

RNA-seq.

The most significant issue was the originally underpowered RNA-seq which had only two replicates. This has been repeated with four replicates now. This has not led to changes in the interpretation of the data between the original study and this one. One comment that the authors make in the response to this was : "Given the robustness of the stage-specific transcriptome, and the legal constraints associated with the use of animals, we chose to limit the number of replicates to the necessary". Ensuring that animal experiments are properly powered and that maximum robustness of the data from the minimum sample size is an important part of experimental design for ethical use of animal models. Essentially the replication here could have been avoided if the original study had used 1 more animal. However, the new version of RNA-seq brings appropriate confidence to the interpretation of the data.

Phosphoproteomics.

The authors provide a robust justification of their strategy for the phosphoproteomics and highlight the inclusion criteria for phosphosites: "Phosphosites were only considered if detected with high confidence (identification FDR<1%) and high localisation confidence (localisation probability >0.75) in at least one replicate". The way missing values were dealt with is explained "For statistical analyses, missing values within a given condition were imputed with a well-established algorithm (MLE) only when at least one observed value was present in that condition." This fills in some of the gaps I was missing from the original manuscript, and I am satisfied that the data analysis is entirely appropriate for a discovery/system -based approach such as this one. The authors also edit the manuscript to reflect that "occupancy" or "stoichiometry" might not be the best description of what they were presenting and switched to the terminology of "normalised phosphorylation level" - I think this is an appropriate response.

Overall, in the absence of follow up experiments on specific individual examples, some of the claims in the original submission were toned down and reflect a more neutral description of the data now. Significantly, the data still underpin a key role for regulation of the ribosome between the amastigote and promastigote stages (and during the differentiation process). The recursive and reciprocal links between the phosphorylation and ubiquitination systems are interesting and present many opportunities for future investigation.

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Reviewer #3 (Public review):

Summary:

The authors proposed to use 5-layer systems level analysis (genomics, transcriptomics, proteomics / protein degradation, metabolomics, phosphoproteomics) to uncover how post-transcriptional mechanisms regulate stage differentiation in *Leishmania donovani*.

This enabled the identification of several potential regulatory networks, including the regulation of stage-specific gene clusters by RNA stabilisation or decay, proteasomal degradation and protein phosphorylation.

In the new version of this manuscript, the authors have addressed all questions raised by the reviewers.

Strengths:

Although some observations in this study have already been described in the literature, the integrated analysis applied here provides a novel view on how different levels of post-transcriptional networks regulate *Leishmania* differentiation. This "5-layer system" represents the first analysis of this depth in kinetoplastid parasites.

The revised version with an increased sample number for the RNA-seq now made the authors assumptions adequate to their obtained data.

The use of a proteasomal inhibitor adds an interesting insight in how protein degradation is involved in the parasite differentiation, confirming previous observations in the literature, and help to explain the discrepancies between mRNA and protein expression in the different stages.

Weaknesses:

While this work provides an impressive and foundational dataset, it opens the door for future research to rigorously validate these initial findings and conclusions.

Significance and Impact in the field.

The different datasets generated in this study will be of great interest to the parasitology community, either to be used for hypothesis generation, to validate data from other sources, etc.

The multi-layered analysis performed here identified a series of potential feedback loops and regulatory networks to be further explored in organisms that lack transcriptional control.

<https://doi.org/10.7554/eLife.111115.1.sa1>

Author response:

General Statements

We thank the reviewers for their insightful and constructive comments, which have substantially strengthened the manuscript. We have addressed all concerns and replaced the previous nonquantitative RNA-seq analysis with a new analysis that allowed for quantitative assessment. We were encouraged to find that the revised analysis not only confirmed our original observations but also reinforced and extended our conclusions.

Point-by-point description of the revisions

Reviewer #1:

Significance

*At its current stage, this work represents a robust resource for molecular parasitology research programs, paving the way for mechanistic studies on multilayered gene expression control and it would benefit from experimental evidence for some of the claims concerning the in silico regulatory networks. Terms like "regulons", "recursive feedback loop" are employed without solid confirmation or extensive literature support. In my view, the most relevant contribution of this study is centered in the direct association between proteasome-dependent degradation and *Leishmania* differentiation.*

We thank the reviewer to acknowledge the impact of our work as a robust resource for further mechanistic studies. We agree that the new concepts emerging from our multilayered analysis should be experimentally assessed. However, given the scope of our analysis (i.e. a complete systems-level analysis of bona fide, hamster-isolated *L. donovani* amastigotes and derived promastigotes) and the amount of data presented in the current manuscript, such functional genetic analysis will merit an independent, in-depth investigation. The current version has been very much toned down and modified to emphasize the impact of our work as a powerful new resource for downstream functional analyses.

Evidence, reproducibility and clarity

The narrative becomes somewhat diffuse with the shift to putative multilevel regulatory networks, which would benefit from further experimental validation.

We agree with the reviewer and toned down the general discussion while suggesting putative multilevel regulatory networks for follow-up, mechanistic analyses. We now emphasize those networks for which evidence in trypanosomatids and other organisms has been published. Experimental validation of some of these regulatory networks is outside the scope of our manuscript and will be pursued as part of independent investigations.

Major issues

Fig.1D suggests a significant portion of the SNPs are exclusive, with a frequency of zero in one of the two stages. Were only the heterozygous and minor alleles plotted in Fig.1D, since frequencies close to 1 are barely observed? Is the same true in Sup Fig. S2B? Why do chrs 4 and 33 show unusual patterns in S2B?

We thank the reviewer for this observation. The SNPs exclusive to either one or the other stage are likely the result of the 10% cutoff we use for this kind of analysis (eliminating SNPs that lack sufficient support, i.e. less than 10 reads). Due to bottle neck events (such as in vitro culture or stage differentiation), many low frequency SNPs are either 'lost' (filtered out) or 'gained' (passing the 10% cutoff) between the ama and pro samples. All SNPs above 10% were plotted. The absence of SNPs at 100% is one of the hallmarks of the Ld1S *L. donovani* strain we are using. Instead, these parasites show a majority of SNPs at a frequency of around 50%, which is likely a sign of a previous hybridization event. Chr 4 and chr 33 show a very low SNP density, most likely as they went through a transient monosomy at one moment of their evolutionary history, causing loss of heterozygosity. We now explain these facts in the figure legend.

Chr26 revealed a striking contrasting gene coverage between H-1 and the other two samples. While a peak is observed for H-1 in the middle of this chr, the other two show a decrease in coverage. Is there any correlation with the transcriptomic/proteomic findings?

This analysis is based on normalized median read depth, taking some variations into account. This is now more clearly specified in the figure legend. We do not see any significant expression changes that would correlate with the observed (minor) read depth changes. As indicated in the legend, we do not consider such small fluctuations (less than ± 1.5 fold) as significant. The reversal of the signal for chr 26 sample H1 eludes us (but again, these fluctuations are minor and not observed at mRNA level).

The term "regulon" is used somewhat loosely in many parts of the text. Evidence of co-transcriptomic patterns alone does not necessarily demonstrate control by a common regulator (e.g., RNA-binding protein), and therefore does not fulfill the strict definition of a regulon. It should be clear whether the authors are highlighting potential multiple

inferred regulons within a list of genes or not. Maybe functional/ gene module/cluster would be more appropriate terms.

We thank the reviewer for this important comment. We replaced ‘regulon’ throughout the manuscript by ‘co-regulated, functional gene clusters’ (or similar).

It is unclear whether the findings in Fig.3E are based on previous analysis of stagespecific rRNA modifications or inferred from the pre-snoRNA transcriptomic data in the current work or something else. I struggle to find the significance of presenting this here.

We thank the reviewer for this comment. Yes, these data show stage-specific rRNA modifications based on previous analyses that mapped stage-specific differences of pseudouridine (Y) (Rajan et al., Cell Reports 2023, DOI: 10.1016/j.celrep.2024.114203) and 2'O-modifications (Rajan et al., Nature Com, in revision) by various RNA-seq analyses and cryoEM. This figure has been modified in the revised version to consider the identification of stageregulated snoRNAs in our new and statistically robust RNA-seq analysis. These data are shown to further support the existence of stage-regulated ribosomes that may control mRNA translatability, as suggested by the enriched GO terms ‘ribosome biogenesis’, ‘rRNA processing’ and ‘RNA methylation’ shown in Figure 2. We better integrated these analyses by moving the panels from Figure 3 to Figure 2.

The protein turnover analysis is missing the critical confirmation of the expected lactacystin activity on the proteasome in both ama and pro. A straightforward experiment would be an anti-polyUb western blotting using a low concentration SDS-PAGE or a proteasome activity assay on total extracts.

We thank the reviewer for this comment and have now included an anti-polyUb Western blot analysis (see Fig S7).

The viability tests upon lactacystin treatment need a positive control for the PI and the YoPro staining (i.e., permeabilized or heat-killed promastigotes).

This control is now included in Fig S7 and we have added the corresponding description to the text.

I found that the section on regulatory networks was somewhat speculative and less focused. Several of the associated conclusions are, in some parts, overstated, such as in "uncovered a similar recursive feedback loop" (line 566) or "unprecedented insight into the regulatory landscape" (line 643). It would be important to provide some form of direct evidence supporting a functional connection between phosphorylation/ubiquitination, ribosome biogenesis/proteins and gene expression regulation.

We agree with the reviewer and have considerably toned down our statements. Functional analyses to investigate and validate some of the shown network interactions are planned for the near future and will be published separately.

Minor issues

(1) The ordinal transition words "First,"/"Second," are used too frequently in explanatory sections. I noted six instances. I suggest replacing or rephrasing some to improve flow.

Rectified, thanks for pointing this out.

(2) Ln 168: Unformatted citations were given for the Python packages used in the study.

Rectified, thanks for pointing this out.

(3) Fig.1D: "SNP frequency" is the preferred term in English.

Corrected.

(4) Fig.2A: not sure what "counts" mean.

This figure has been replaced.

(5) Ln 685: "Transcripts with $FC < 2$ and adjusted p -value > 0.01 are represented by black dots" > This sentence is inaccurate. The intended wording might be: "Transcripts with $FC < 2$ OR adjusted p -value > 0.01 are represented by black dots"

We thank the reviewer and corrected accordingly.

(6) Ln 698: Same as Ln 685 mentioned above.

We thank the reviewer and corrected accordingly.

(7) Fig.2B and elsewhere: The legend key for the GO term enrichment is a bit confusing. It seems like the color scales represent the adj. p -values, but the legend keys read "Cluster efficiency" and "Enrichment score", while those values are actually represented by each bar length. Does light blue correspond to a max value of 0.05 in one scale, and dark blue to a max value of 10^{-7} in the other scale?

This was corrected in the figure and the legends were updated accordingly.

(8) Sup Figure S3A and S4A: The hierarchical clustering dendrograms are barely visible in the heatmaps.

Thanks for the comment. Figure S3 was removed and replaced by a hierarchical clustering and a PCA plot.

(9) S3A Legend: The following sentence sounds a bit awkward: "Rows and columns have been re-ordered thanks to a hierarchical clustering". I suggest switching "thanks to a hierarchical clustering" to "based on hierarchical clustering".

This figure was removed and the legend modified.

(10) Fig.5D: The font size everywhere except the legend key is too small. In addition, on the left panel, gene product names are given as a column, while on the right, the names are shown below the GeneIDs. Consistency would make it clearer.

Thank you, this is now rectified. To ensue readability, we reduced the number of shown protein kinase examples.

Reviewer #2 Evidence, reproducibility and clarity:

In the absence of riboprofiling the authors return to the RNA-seq to assess the levels of pre-Sno RNA (the role of the could be more explicitly stated).

We thank the reviewer for this comment. We moved the snoRNA analysis from Fig 3 to Fig 2 (see also the similar comment of reviewer 1), which better integrates and justifies this analysis. Based on the new and statistically robust RNA-seq analysis, the volcano plot showing differential snoRNA expression and possible ribosome modification has been adjusted (Figures 2C and D).

The authors provide a clear and comprehensive description of the data at each stage of the results and this is woven together in the discussion allowing hypotheses to be formed on the potential regulatory and signalling pathways that control the differentiation of

amastigotes to promastigotes. Given the amount and breadth of data presented the authors are able to present a high-level assessment of the processes that form feedback loops and/or intersectional signalling, but specific examples are not picked out for deeper validation or exploration.

We thank the reviewer to acknowledge the amount and breadth of data presented. As indicated above (see responses to reviewer 1), mechanistic studies will be conducted in the near future to validate some of the regulatory interactions. These will be subject of separate publications. As noted above (response to reviewer 1), we toned down the general discussion, suggest follow-up mechanistic analyses and emphasize those networks for which evidence in trypanosomatids and other organisms has been published.

Major comments:

(1) As I have understood it from the description in the text, and in Data Table 4, the RNA-seq element of the work has only been conducted using two replicates. If this is the case, it would substantially undermine the RNA-seq and the inferences drawn from it. Minimum replicates required for inferential analysis is 3 bio-replicates and potentially up to 6 or 12. It may be necessary for the authors to repeat this for the RNA-seq to carry enough weight to support their arguments. (PMID: 27022035)

We agree with the reviewer and conducted a new RNA-seq analysis with 4 independent biological replicates of spleen-purified amastigotes and derived promastigotes. Given the robustness of the stage-specific transcriptome, and the legal constraints associated with the use of animals, we chose to limit the number of replicates to the necessary. We thank the reviewer for this important comment, and the new data not only confirm the previous one (providing a high level of robustness to our data) but allowed us to increase the number of identified stage-regulated snoRNAs, thus further supporting a possible role of ribosome modification in *Leishmania* stage development.

(2) There are several examples that are given as reciprocal or recursive signalling pathways, but these are not followed up with independent, orthogonal techniques. I think the paper currently forms a great resource to pursue these interesting signalling interactions and is certainly more than just a catalogue of modifications, but to take it to the next level ideally a novel signalling interaction would be demonstrated using an orthogonal approach. Perhaps the regulation of the ribosomes could have been explored further (same teams recently published related work on this). Or perhaps more interestingly, a novel target(s) from the ubiquitinated protein kinases could have been explored further; for example making precision mutants that lack the ubiquitination or phosphorylation sites - does this abrogate differentiation?

We agree with the reviewer that the paper currently forms a great resource. In-depth molecular analysis investigating key signaling pathways and regulatory interactions are outside the scope of the current multilevel systems analysis but will be pursued in independent investigations.

(3) I found the use of lactacystin a bit curious as there are more potent and specific inhibitors of Leishmania proteasomes e.g. LXE-408. This could be clarified in the write-up (See below).

We thank the reviewer for this comment. We opted for the highly specific and irreversible proteasome inhibitor lactacystin that has been previously applied to study the *Leishmania* proteasome (PMID: 15234661) rather than the trypanosomatid-specific drug candidate LXE408 as the strong cytotoxic effect of the latter makes it difficult to distinguish between direct effects on protein turnover and secondary effects resulting from cell death, limiting its utility for dissecting proteasome function in living parasites. We have added this information in the Results section.

(4) If it is the case that only 2 replicates of the RNA-Seq have been performed it really is not the accepted level of replication for the field. Most studies use a minimum of 3 bioreplicates and even a minimum of 6 is recommended by independent assessment of DESeq2.

See response to comment 1 above.

(5) As far as I could see, the cell viability assay does not include a positive control that shows it is capable of detecting cytotoxic effects of inhibitors. Add treatment showing that it can differentiate cytostatic vs cytotoxic compound.

This control has now been added to Fig S7.

(6) It is realistic for the authors to validate the cell viability assay. If the RNA-seq needs to be repeated then this would be a substantial involvement.

Redoing the RNA-seq analysis was entirely feasible and very much improved the robustness of our results.

(7) All the methods are written to a good level of detail. The sample prep, acquisition and data analysis of the protein mass spectrometry contained a high level of detail in a supplemental section. The authors should be more explicit about the amount of replication at each stage, as in parts of the manuscript this was quite unclear.

We thank the reviewer for this comment and explicitly state the number of replicates in Methods, Results and Figure legends for all analyses. The number of replicates for each analysis is further shown in the overview Figure S1.

(8) Unless I have misunderstood the manuscript, I believe the RNA-seq dataset is underpowered according to the number of replicates the authors report in the text.

See response to comment 1 above.

(9) Looking at Figure 1 and S1 and Data Table 4 to show the sample workflow I was surprised to see that the RNA-seq only used 2 replicates. The authors do show concordance between the individual biological replicates, but I would consider that only having 2 is problematic here, especially given the importance placed on the mRNA levels and linkage in this study. This would constitute a major weakness of the study, given that it is the basis for a crucial comparison between the RNA and protein levels.

We agree and have repeated the RNAseq analysis using four independent biological replicates - see response to comment 1.

(10) It also wasn't clear to me how many replicates were performed at each condition for the lactacystin treatment experiment - can the authors please state this clearly in the text, it looks like 4 replicates from Figure S1 and Data Table 8.

Indeed, we did 4 replicates. This is now clarified in Methods, Results and Figure legends and shown in Figure S1.

(11) Four replicates are used for the phosphoproteomics data set, which is probably ok, but other researchers have used a minimum of 5 in phosphoproteomics experiments to deal with the high level of variability that can often be observed with low abundance proteins & modifications. The method for the phosphoproteomics analysis suggests that a detection of a phosphosite in 1 sample (also with a localisation probability of >0.75) was required for then using missing value imputation of other samples. This seems like a low threshold for inclusion of that phosphosite for further relative quantitative analysis. For example, Geoghegan et al (2022) (PMID: 36437406) used a much more stringent threshold of greater than or equal to 2 missing values from 5 replicates as an exclusion criteria for detected phosphopeptides. Please correct me if I misunderstood the data processing, but as it stands the imputation of so many missing values (potentially 3 of 4 per sample category) could be reducing the quality of this analysis.

We thank the reviewer for this remark and for highlighting best practices in phosphoproteomics data analysis. Unlike other studies that use cultured parasites and thus have access to unlimited amounts, our study employs bona fide amastigotes isolated from infected hamster spleens. In France, the use of animals is tightly controlled and only the minimal number of animals to obtain statistically significant results is tolerated (and necessary to obtain permission to conduct animal experiments).

Regarding the number of biological replicates, we would like to emphasize that the use of four biological replicates is fully acceptable and used in quantitative proteomics and phosphoproteomics, particularly when combined with high-quality LC-MS/MS data and stringent peptide-level filtering. While some studies indeed employ five or more replicates, this is not a strict requirement, and many high-impact phosphoproteomics studies have successfully relied on four replicates when experimental quality and depth are high. In the present study, we adopted a discovery-oriented approach, aimed at detecting as many confidently identified phosphopeptides as possible. The consistency between replicates, combined with the depth of coverage and signal quality, indicates that four replicates are adequate for both the global proteome and the phosphoproteome in this context. Importantly, the quality of the MS data in this study is supported by (i) a high number of confidently identified peptides and phosphopeptides (identification FDR $<1\%$), (ii) robust phosphosite localisation probabilities (localisation probability >0.75), and (iii) reproducible quantitative profiles across replicates. Notably, most of the identified phosphopeptides are quantified in at least two replicates within a given condition (between 73.2% and 83.4% of all the identified phosphopeptides among replicates of the same condition).

Regarding missing value imputation, we appreciate that our initial description may have been unclear and we have revised the Methods to avoid misunderstanding. Phosphosites were only considered if detected with high confidence (identification FDR $<1\%$) and high localisation confidence (localisation probability >0.75) in at least one replicate. This criterion was chosen to retain biologically relevant, low-abundance phosphosites, which are more difficult to identify and are often stochastically sampled in phosphoproteomics datasets. For statistical analyses, missing values within a given condition were imputed with a well-established algorithm (MLE) only when at least one observed value was present in that condition. Notably, they were replaced by values in the neighborhood of the observed intensities, rather than by globally low, noise-like values.

We agree that more stringent exclusion rules, such as those used by Geoghegan et al. (2022), are appropriate in some contexts. However, there is no universally accepted standard for missingness thresholds in phosphoproteomics, and different strategies reflect trade-offs between sensitivity and stringency. In our discovery-oriented approach, we deliberately prioritized biological coverage while maintaining data quality. Our main conclusions are supported by coherent biological patterns, rather than by isolated phosphosite measurements.

(12) For the metabolomics analysis it looks like 2 amastigote samples were compared against 4 promastigote samples. Why not triplicates of each?

We thank the reviewer for noticing this point. It is an error in the figure file (Sup figure S1). Four biological replicates of splenic amastigotes were prepared (H130-1, H130-2, H133-1 and H133-2). Amastigotes from 2 biological replicates (H131-1 and H131-2) were seeded for differentiation into promastigotes in 4 flasks (2 per biological replicate) that were collected at passage 2. We have updated the figure file accordingly.

Minor comments:

Are prior studies referenced appropriately?

Yes

Are the text and figures clear and accurate?

The write up is clear, with the data presented coherently for each method. The analyses that link everything together are well discussed. The figures are mostly clear (see below) and are well described in the legends. There is good use of graphics to explain the experimental designs and sample names - although it is unclear if technical replicates are defined in these figures.

We thank the reviewer for these positive comments. We now included the information on replicates in the overview figure (Figure S1).

As I have understood it, the authors have calculated the "phosphostoichiometry" using the ratio of change in the phosphopeptide to the ratio of the change in total protein level changes. This is detailed in the supplemental method (see below). Whilst this has normalised the data, it has not resulted in an occupancy or stoichiometry measurement, which are measured between 0-1 (0% to 100%). The normalisation has probably been sufficient and useful for this analysis, but this section needs to be re-worded to be more precise about what the authors are doing and presenting. These concepts are nicely reviewed by Muneer, Chen & Chen 2025 (PMID: 39696887) who reference seminal papers on determination of phosphopeptide occupancy - and may be a good place to start. An alternative phrase should be used to describe the ratio of ratios calculated here, not phosphostoichiometry.

We thank the reviewer for this insightful comment and fully agree with the conceptual distinction raised. The reviewer is correct that the approach used in this study does not measure absolute phosphosite occupancy or stoichiometry, which would indeed require dedicated experimental strategies and would yield values bounded between 0 and 1 (0–100%). Instead, we calculated a normalized phosphorylation change, defined as the ratio of the change in phosphopeptide abundance relative to the change in the corresponding total protein abundance (a ratio-of-ratios approach – see doi:10.1007/978-1-0716-1967-4_12), and we tested whether this normalized phosphorylation change differed significantly from zero. This normalization approach is comparable to those previously published in the «

Experimental Design and Statistical Analysis of the Proteome and the Phosphoproteome » section of the following paper (DOI: 10.1016/j.mcpro.2022.100428).

Our intention was to account for protein-level regulation and thereby better isolate changes in phosphorylation dynamics. While this normalization is informative and appropriate for the biological questions addressed here, we agree that the term “phosphostochiometry” is imprecise and not correct in this context.

In response, we (i) replaced the term “phosphostochiometry” throughout the manuscript with a more accurate description, such as “normalized phosphorylation level”, or “relative phosphorylation change normalized to protein abundance”, and (ii) revised the corresponding Methods and Results text to clearly state that absolute occupancy was not measured.

This rewording will improve conceptual accuracy without altering the validity or interpretation of the results.

From the authors methods describing the ratio comparison approach: "Another statistical test was performed in a second step: a contrasted t-test was performed to compare the variation in abundance of each modified peptide to the one of its parent unmodified protein using the limma R package {Ritchie, 2015; Smyth, 2005}. This second test allows determining whether the fold-change of a phosphorylated peptide between two conditions is significantly different from the one of its parent and unmodified protein (paragraph 3.9 in Gai Gianetto et al 2023). An adaptive Benjamini-Hochberg procedure was applied on the resulting pvalues thanks to the adjust.p function of R package cp4p {Gai Gianetto, 2016} using the Pounds et al {Pounds, 2006} method to control the False Discovery Rate level."

The references have been formatted.

Several aspects of the figures that contain STRING networks are quite useful, particularly the way colour around the circle of each node to denote different molecular functions/biological processes. However, some have descended into "hairball" plots that convey little useful information that would be equally conveyed in a table, for example. Added to this, the points on the figure are identified by gene IDs which, while clear and incontrovertible, are lacking human readability. I suggest that protein name could be included here too.

We thank the reviewer for this comment but for readability we opted to keep the figure as is. We now refer to Tables 8, 9, and 12 that allow the reader to link gene IDs to protein name and annotation (if available).

It is also not clear what STRING data is being plotted here, what are the edges indicating - physical interactions proven in Leishmania, or inferred interactions mapped on from other organisms? Perhaps as supplemental data provide the Cytoscape network files so readers can explore the networks themselves?

We thank the reviewer for this comment. While the STRING plugin in Cytoscape enables integrated network-based analyses, it represents protein–protein associations as a single edge per protein pair derived from the combined confidence score. Consequently, the specific contribution of individual evidence channels (e.g. experimental evidence, curated databases, coexpression, or text mining) cannot be disentangled within this framework. However, this representation was considered appropriate for the present study, which focused on global network topology and functional enrichment rather than on the interpretation of individual interaction types. The information on stringency has been added to the Methods section and the Figure legends (adding the information on confidence score cutoff).

We decided not to submit the Cytoscape files as they were generated with previous versions of Cytoscape and the STRING plugin. Based on the differential abundance data shown in the tables it will be very easy to recreate these networks with the new versions for any follow up study.

The title of columns in table S10 panel A are written in French, which will be ok for many people particularly those familiar with proteomics software outputs, but everything else is in English so perhaps those titles could be made consistent.

We apologize and have translated the text in English.

I would suggest that the authors provide a table that has all the gene IDs of the Ld152D strain and the orthologs for at least one other species that is in TriTrypDB. This would make it easy to interrogate the data and make it a more useful resource for the community who work on different strains and species of Leishmania. Although this data is available it is a supplemental material file in a previous paper (Bussotti et al PNAS 2021) and not easy to find.

We thank the reviewer for this very useful suggestion and have added this table (Table S13).

Figure 5b - from the legend it is not clear where the confidence values were derived in this analysis, although this is explained in the supplemental method. Perhaps the legend can be a bit clearer.

We have the following statement to the legend: ‘Confidence values were derived as described in Supplementary Methods’.

Can the authors discuss why lactacystin was used? While this is a commonly used proteasome inhibitor in mammalian cells there is concern that it can inhibit other proteases. At the concentrations (10 μ M) the authors used there are off-target effects in Leishmania, certainly the inhibition of a carboxypeptidase (PMID: 35910377) and potentially cathepsins as is observed in other systems (PMID: 9175783). There is a specific inhibitor of the Leishmania proteasome LXE-408 (PMID: 32667203), which comes closer to fulfilling the SGC criteria (PMID: 26196764) for a chemical probe - why not use this. Does lactacystin inhibit a different aspect of proteasome activity compared to LXE-408?

We have add the following justification to the results section (see also response above to comment 3 for reviewer 2): We chose the highly specific and irreversible proteasome inhibitor lactacystin over the trypanosomatid-specific, reversible drug candidate LXE408 as the latter’s potent cytotoxicity can confound direct effects on protein turnover with secondary consequences of cell death, limiting its utility for dissecting proteasome function in living parasites.

The application of lactacystin is changing the abundance of a multitude of proteins but no precision follow up is done to identify if those proteins are necessary and/or sufficient from driving/blocking differentiation. This could be tested using precision edited lines that are unable to be ubiquitinated? There is a lack of direct evidence that the proteins protected from degradation by lactacystin are ubiquitinated? Perhaps some of these could be tagged and IP'd then probed for ubiquitin signal. Di-Gly proteomics to reveal ubiquitinated proteins? These suggestions should be considered as OPTIONAL experiments in the relevant section above.

We very much appreciate these very interesting suggestions, which we will be considered for ongoing follow-up studies.

*In the data availability RNA-seq section the text for the GEO link is :
(<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE227637>) but the embedded*

link takes me to (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165615>) which is data for another, different study. Also, the link to the GEO site for the DNA seq isn't working and manual searches with the archive number (BioProject PRJNA1231373) does not appear to find anything. The IDs for the mass spec data PRIDE/ProteomeXchange don't seem to bring up available datasets: PXD035697 and PXD035698

The links have now been rectified and validated. For those data that are still under quarantine, here is the login information: To access the data:

DNaseq data: <https://dataview.ncbi.nlm.nih.gov/object/PRJNA1231373?reviewer=6qt24dd7f475838rbqfn228d> 0

RNAseq data: <https://www.ebi.ac.uk/biostudies/ArrayExpress/studies/E-MTAB-16528?key=65367b55-d77f4c06-b4bd-bc10f2dc0b14>

Proteomic data: <http://www.ebi.ac.uk/pride>

Phosphoproteomic data: <http://www.ebi.ac.uk/pride>

Significance

Strengths:

(1) The molecular pathways that regulate Leishmania life-stage transitions are still poorly understood, with many approaches exploring single proteins/RNAs etc in a reductionist manner. This paper takes a systems-scale approach and does a good job of integrating the disparate -omics datasets to generate hypotheses of the intersections of regulatory proteins that are associated with life-cycle progression.

We thank the reviewer for this positive assessment of our work.

(2) The differentiation step studied is from amastigote to promastigote. I am not aware that this has been studied before using phosphoproteomics. The use of the hamster derived amastigotes is a major strength. While a difficult/less common model, the use of hamsters permits the extraction of parasites that are host adapted and represent "normal", host-adapted Leishmania ploidy, the promastigote experiments are performed at a low passage number. This is a strength of the work as it reduces the interference of the biological plasticity of Leishmania when it is cultured outside the host.

We thank the reviewer for the acknowledgment of our relevant hamster system, for which we face many challenges (financial, ethical, administrative as protocols need to be approved by the French government).

Limitations:

Potential lack of appropriate replication (see above).

See response to comment 1.

Lack of follow up/validation of a novel signalling interaction identified from the systems-wide approach. There is a lack of assessment of whether a single signalling cascade is driving the differentiation or these are all parallel, requisite pathways. The authors state the differentiation is not driven by a single master regulator, but I am not sure there is adequate evidence to rule this in or out.

See response to comment 2 above.

The study applies well established techniques without any particular technical stepchange. The application of large-scale multi-omics techniques and integrated comparisons of the different experimental workflows allow a synthesis of data that is a step forward from that existing in the previous Leishmania literature. It allows the generation of new hypotheses about specific regulatory pathways and crosstalk that potentially drive, or are at least active, during amastigote>promastigote differentiation.

We thank the reviewer for these positive comments.

This manuscript will have primary interest to those researchers studying the molecular and cell biology of Leishmania and other kinetoplastid parasites. The approaches used are quite standard (so not so interesting in terms of methods development etc.) and given the specific quirks of Leishmania biology it may not be that relevant to those working more broadly in parasites from different clades/phyla, or those working on opisthokont systems- yeast, humans etc. Other Leishmania focused groups will surely cherry-pick interesting hits from this dataset to advance their studies, so this dataset will form a valuable reference point for hypothesis generation.

We thank the reviewer for this assessment and agree that our data sets will be very valuable for us and other teams to generate hypotheses for follow-up studies.

Relevant expertise: Trypanosoma & Leishmania molecular & cell biology, RNA-seq, proteomics, transcriptional/epigenetic regulation, protein kinases - some experience of UPS system.

I have not provided comment on the metabolomics as it is outside my core expertise. However, I can see it was performed at one of the leading parasitology metabolomics labs.

We thank the reviewer for sharing expertise, investing time and intelligence in the assessment of our manuscript, and the highly constructive criticisms provided.

Reviewer #3 (Evidence, reproducibility and clarity):

Summary:

The study presents a comprehensive multi-omics investigation of Leishmania differentiation, combining genomic, transcriptomic, proteomic, phospho-proteomic and metabolomic data. The authors aim to uncover mechanisms of post-transcriptional and post-translational regulation that drive the stage-specific biology of L. donovani. The authors provide a detailed characterization of transcriptomic, proteomic, and phospho-proteomic changes between life stages, and dissect the relative contributions of mRNA abundance and protein degradation to stage-specific protein expression. Notably, the study is accompanied by comprehensive supplementary materials for each molecular layer and provides public access to both raw and processed data, enhancing transparency and reproducibility. While the data are rich and compelling, several mechanistic interpretations (e.g., "feedback loops," "recursive networks," "signaling cascades") are overstated. Similarly, the classification of gene sets as "regulons" is not adequately supported, as no common regulatory factor has been identified and only a single condition change (amastigote to promastigote) was assessed.

We thank the reviewer for these comments and have corrected the manuscript to eliminate all unjustified mechanistic interpretations.

Major Comments:

(1) Across several sections (incl abstract, L559-565, L589-599, L600-L603, L610-612, L613-614, L625, L643-645, L650-652), the manuscript describes "recursive or self-controlling networks", "signaling cascades", "self-regulating", and "recursive feedback loops" - involving protein kinases, phosphatases, and translational regulators. While the data convincingly demonstrate stage-specific changes in phosphorylation and abundance changes in key molecules, the language used implies causal, direct and directional regulatory relationships that have not been experimentally validated.

We agree with the reviewer and have corrected the text, replacing all expressions that may allude to causal or directional relationships by more neutral expressions such as 'coexpression'.

(2) Co-expression and shared function alone do not define a regulon (L363, and several other places in the manuscript). A regulon also requires the gene set to be regulated by the same factor, for which there is no evidence here. Regulons can be derived from transcriptomic experiments, but then they need to show the same transcriptional behavior across many biological conditions, while here just 1 condition change is evaluated. Therefore, this analysis is conventional GO enrichment analysis and should not be overinterpreted into regulons.

We agree with the reviewer and have replaced 'regulon' with 'co-regulated gene clusters' (or similar).

(3) LFQ intensity of 0 (e.g., L389): An LFQ intensity of 0 does not necessarily indicate that a protein is absent, but rather that it was not detected. This can occur for several reasons: (1) true biological absence in one condition, (2) low abundance below the detection threshold, or (3) stochastic missingness due to random dropout in mass spectrometry. While the authors state that adjusted p-values for the 1534 proteins exclusively detected in either amastigotes or promastigotes are below 0.01, I could not find corresponding p-values for these proteins in Table 8 ('Global_Proteomic'). An appropriate statistical method designed to handle this type of missingness should be used. In this context, I also find the following statement unclear: "identified over 4000 proteins at each stage in at least 3 out of 4 biological replicates, representing 3521 differentially expressed proteins (adjusted p-value < 0.01), 1534 of which were exclusively detected in either ama or pro." If a protein is exclusively detected in one stage, then by definition it should not be detected in that number of replicates at both stages. This apparent contradiction should be clarified.

We fully agree with the reviewer, an LFQ intensity of 0 may results from various reasons. We realize that our wording may have been ambiguous. For clarity, we have modified the original text to: 'Label-free quantitative proteomic analysis of 4 replicates of amastigotes and derived promastigotes identified over 4000 proteins, including 1987 differentially expressed proteins (adjusted p-value < 0.01), and 1534 that were exclusively detected in either ama or pro (Figure 3A left panel, Table 6).' We also modified the legend of the Figure 3B. Concerning missing values that could be either missing not at random (MNAR) or missing completely at random (MCAR), rather than introducing potentially misleading imputed values, we chose to treat these missing values as genuine stage-specific differences (presence/absence): quantitative statistics are restricted to proteins with measurable LFQ in both stages, while proteins with consistent presence in one stage and non-detection in the other are reported as stage-restricted detections. We believe this strategy is transparent and minimizes modeling assumptions, while still highlighting robust stage-specific signals. Our approach is supported by independent validation through RNA-seq data, which corroborates the differential presence/absence patterns observed at the protein level. Furthermore, our enrichment analyses reveal significant over-representation of specific biological terms among these stage-specific proteins, providing biological coherence to these findings. Therefore, we

believe our conservative approach of treating these as genuine presence/absence differences, validated by orthogonal data, is more appropriate than introducing imputed values based on arbitrary statistical assumptions.

(4) L412 - Figure 3B: The figure shows proteins with infinite fold changes, which result from division by zero due to LFQ intensity values of zero in one of the compared conditions. As previously noted, interpreting LFQ zero values as true absence of expression is problematic, since these zeros can arise from several technical reasons - such as proteins being just below the detection threshold or due to stochastic dropout during MS analysis. Therefore, the calculated fold changes for these proteins are likely highly overestimated. This concern is visually supported by the large gap on the y-axis (even in log scale) between these "infinite" fold changes and the rest of the data. Moreover, given Leishmania's model of constitutive gene expression, it seems biologically implausible that all these proteins would be completely absent in one stage. This issue applies not only to Figure 3B, but also to the analyses presented in Figures 4D and 4E.

We thank the reviewer for this comment. To clarify this section, we modified the text as follows: ‘Only expression changes were considered that either showed statistically significant differential abundance at both RNA and protein levels ($p < 0.01$), or showed significant RNA changes ($p < 0.01$) with the corresponding protein being detected in only one of the two stages. These latter proteins are identified by signals that were arbitrarily placed at the upper (detected in ama) or the lower (detected in pro) parts of the graph. Whether these proteins just escape detection due to low expression or are truly not expressed remains to be established.’ We also deleted the ‘infinity’ symbol from the Figure.

Minor Comments:

Methods

L132: Typo: "A according" should be "according."

The ‘A’ refers to RNase A. We added a comma for clarification (...RNase A, according to...)

L158: How exactly were somy levels calculated? Please specify the method used, as I could not find a clear description in the referenced manuscript.

We thank the reviewer for this comment. Aside the already quite detailed description in Methods and the reference there to the paper describing the pipeline, we now added a link to the description of the karyotype module of the giptools package (<https://gip.readthedocs.io/en/latest/giptools/karyotype.html>). There the following explanation can be found: “The karyotype module aims at comparing the chromosome sequencing coverage distributions of multiple samples. This module is useful when trying to detect chromosome ploidy differences in different isolates. For each sample the module loads the GIP files with the bin sequencing coverage (.covPerBin.gz files) and normalizes the meancoverage values by the median coverage of all bins. The bin scores are then converted to somy scores which are then used for producing plots and statistics.” The description then goes into further detail.

L158: Chromosome 36 is not consistently disomic, as stated. It has been observed in other somy states (e.g., Negreira et al. 2023, EMBO Reports, Figure 1), even if such occurrences are rare in the studied context. Normalizing by chr36 remains a reasonable choice, but it would be helpful to confirm that the majority of chromosomes appear disomic post-normalization to support the assumption that chr36 is disomic in this dataset as well.

We thank the reviewer for this comment. Unlike the paper cited above (using longterm cultured promastigotes), our analysis uses promastigote parasites from early culture

adaptation (p2) that were freshly derived from splenic amastigotes known to be disomic (and confirmed here), which represents an internal control validating our analysis.

L163: Suggestion: Cite the GIP pipeline here rather than delaying the reference until L173.

Corrected

L188: "Controlled" may be a miswording. Consider replacing with "confirmed" or "validated."

Corrected to 'validated'

L214: Please specify which statistical test was used to assess differential expression at the protein level. L227: Similarly, clarify which statistical test was applied for determining differential expression in the phospho-proteomics data.

As noted in the Methods section, a limma t-test was applied to determine proteins/phosphoproteins with a significant difference in abundance while imposing a minimal fold change of 2 between the conditions to conclude that they are differentially abundant {Ritchie, 2015; Smyth, 2005}.

Results

L337-339: The interpretation here is too speculative. Phrases like "suggesting" and "likely" are too strong given the evidence presented. Alternative explanations, such as mosaic variation combined with early-stage selective pressure in the culture environment, should be considered.

We thank the reviewers for these suggestions and have reformulated into: 'In the absence of convergent selection, it is impossible to distinguish if these gene CNVs provide some strain-specific advantage or are merely the result of random genetic drift.'

L340: The "undulating pattern" mentioned is somewhat subjective. To support this interpretation, consider adding a moving average (or similar) line to Figure 3A, which would more clearly highlight this trend across the data points.

These lines have been added to Figure 1C (not 3A).

L356: It may be more accurate to say "control of individual gene expression," since Leishmania does have promoters - the key distinction is that initiation does not occur on a gene-by-gene basis.

Corrected

L403-405: The statement "this is because these metabolites comprise a glycosomal succinate shunt..." should be rephrased as a hypothesis rather than a definitive explanation, as this causal link has not been experimentally validated.

Thank you for the comment – we followed your advice.

L407: Replace "confirming" with "matching" to avoid overstating the agreement with previous observations.

Corrected

L408: Replace "correlated" with "matched" for more accurate interpretation of results.

Corrected

L433: It is unclear how differential RNA modifications were detected. Please specify which biological material was used, the number of replicates per life stage, and how statistical evaluation of differential modifications was performed.

This figure has now been updated using our statistically robust RNA-seq analysis conducted for the revision. See comments above.

L436: This conclusion appears incomplete. While the manuscript mentions transcript-regulated proteins, it should also note that other proteins showed discordant mRNA/protein patterns. A more balanced conclusion would mention both the matching and non-matching subsets.

We thank the reviewer for this comment and have made the necessary adjustments to better balance this conclusion.

L441: The phrase "poor correlation" overgeneralizes and lacks nuance. Earlier sections of the manuscript describe hundreds of genes where mRNA and protein levels correlate well, suggesting that mRNA turnover plays a key regulatory role. Please rephrase this sentence to clarify that poor correlation applies only to a subset of the data.

This has been corrected to ‘The discrepancies we observed in a sub-set of genes between....’.

L454: The claim that "epitranscriptomic regulation and stage-adapted ribosomes are key processes" should be supported with references. If this builds on previously published work, please cite it accordingly.

Corrected

L457: Proteasomal degradation is a well-established mechanism in Leishmania. These findings are interesting but should be presented in the context of existing literature (e.g. Silva-Jardim et al.2014, [PMID: 15234661]) rather than as entirely novel.

Corrected

L459: The authors should add a microscopy image of promastigotes treated with lactacystin. This would provide insight into whether treatment affects morphology, as is known in T. cruzi (see Dias et al., 2008). It would be particularly informative if Leishmania behaves differently.

We added this information to Figure S7.

L472 + L481: Table 9 shows several significant GO terms not discussed in the manuscript. Please clarify how the subset presented in the text was selected.

We added this information to the text (‘some of the most significantly enrichment terms included ...’).

L482: The argument that a single master regulator can be excluded is unclear. Could the authors please elaborate on the reasoning or data supporting this conclusion?

This statement was too speculative and has been removed. Instead, we added ‘Thus, Leishmania differentiation correlates with the expression of complex signaling networks that are established in a stage-specific manner’.

L494: The term "unexpected" may not be appropriate here, as protein degradation is a well-established regulatory mechanism in trypanosomatids. Consider omitting this term to better reflect the field's current understanding.

We deleted the term as suggested and reformulated to ‘....our results confirm the important role of protein degradation....’.

L543: The term "feedback loop" should be used more cautiously. The current data are correlative, and no interventional experiments are provided to support a causal regulatory loop between proteasomal activity and protein kinases. As such, this remains a hypothesis rather than a confirmed mechanism.

We fully agree and have toned down the entire manuscript, referring to feedback loops only as a hypothesis and not as a fact emerging from our datasets, which set the stage for future functional analyses.

Discussion

L555: As noted in L494, reconsider using the word "unexpected."

Removed

L589: The data do not fully support the presence of stage-specific ribosomes. Rather, they suggest differential ribosomal function through changes in abundance and regulation. Please consider rephrasing.

We thank the reviewer for this comment and have follow the advice reformulating the sentence according to the suggestion.

L657-658: The discussion of post-transcriptional and post-translational regulation of gene dosage effects would benefit from citing additional literature beyond the authors' own work. E.g. the study by Cuypers et al. (PMID: 36149920) offers a relevant and comprehensive analysis covering 4 'omic layers.

We apologize for this omission and now describe and cite this publication in the Results section when concluding the results shown in Figure 1.

L659-664: The reference to deep learning for biomarker discovery appears speculative and loosely connected to the current findings. As no such methods were applied in the study, and the manuscript does not clarify what types of biomarkers are intended, this statement could be seen as aspirational rather than evidence-based. Consider either omitting or elaborating with clear justification.

We agree and have deleted this section.

L690 + L705 (Figure 2): The phrase "main GO terms" is vague. Please clarify the criteria for selecting the GO terms shown - were they chosen based on adjusted p-value, enrichment score, or another metric? Additionally, define "cluster efficiency," explaining how it was calculated and what it represents.

Corrected to ‘some of the most significantly enriched GO terms’.

Referee cross-commenting

Overall, I think the other reviewers' comments are fair. They seem to align particularly on the following points:

(1) Reviewers agree that this is a comprehensive body of work with original contributions to the field of Leishmania/trypanosomatid molecular biology, and that it will serve as a valuable reference for hypothesis generation.

(2) Several reviewers raise concerns about overinterpretation of the data, particularly regarding regulatory networks, regulons, and master regulators. The interpretation and large parts of the discussion are considered too speculative without additional functional validation.

(3) There are comments about the incorrect statistical treatment of missing values in the proteomics experiments, which affects confidence in some of the conclusions.

(4) While the correlation between the two RNA-Seq replicates is high, the decision to include only two biological replicates is seen as unfortunate and not ideal for statistical robustness.

(5) The use of lactacystin should be more clearly motivated, and its limitations discussed in the context of the experiments.

Even though I did not remark on the last two points (4 and 5) in my own review, I agree with them.

We thank the reviewer for this cross-comparison, which served us as guide to revise our manuscript. We believe that we have responded to all these concerns.

Reviewer #3 (Significance):

This study provides a rich, integrative multi-omics dataset that advances our understanding of stage-specific adaptation in the transcriptionally unique parasite Leishmania. By dissecting the relative contributions of mRNA abundance and protein turnover to final protein levels across life stages, the authors offer valuable insights into post-transcriptional and post-translational regulation. The work represents a resource-driven yet conceptually informative contribution to the field, with comprehensive supplementary materials and transparent data sharing standing out as additional strengths.

However, the mechanistic insights proposed are speculative in several places and require more cautious language. The study is most impactful as a resource and descriptive atlas, initiating hypotheses for future validation. The broad scientific community working on Leishmania, trypanosomatids, and post-transcriptional regulation in eukaryotes would benefit from this work.

We thank the reviewer for this positive assessment and have modified the manuscript to further emphasize its strength as an important resource to incite mechanistic follow-up studies.

Field of reviewer expertise: multi-omics integration, bioinformatics, molecular parasitology, transcriptomics, proteomics, metabolomics, Leishmania, Trypanosoma.

Reviewer #4 (Evidence, reproducibility and clarity):

Summary:

This study investigates the regulatory mechanisms underlying stage differentiation in Leishmania donovani, a parasitic protist. Pesher et al., aim to address the central question of how these parasites establish and maintain distinct life cycle stages in mostly the absence of transcriptional control. The authors employed a five-layered systems-level analysis comparing hamster-derived amastigotes and their in vitro-derived promastigotes. From those parasites, they performed a genomic, transcriptomic, proteomic, metabolomic and phosphoproteomic analysis to reveal the changes the parasites undertook between the two life stages.

The main conclusion stated by the authors are:

- The stage differentiation in vitro is largely independent of major changes in gene dosage or karyotype.

- RNA-seq analysis identified substantial stage-specific differences in transcript abundance, forming distinct regulons with shared functional annotations. Amastigotes showed enrichment in transcripts related to amastins and ribosome biogenesis, while promastigotes exhibited enrichment in transcripts associated with ciliary cell motility, oxidative phosphorylation, and posttranscriptional regulation itself.

- Quantitative phosphoproteome analysis revealed a significant increase in global protein phosphorylation in promastigotes. Normalizing phosphorylation changes against protein abundance identified numerous stage-specific phosphoproteins and phosphosites, indicating that differential phosphorylation also plays a crucial role in establishing stage-specific biological networks. The study identified recursive feedback loops (where components of a pathway regulate themselves) in post-transcriptional regulation, protein translation (potentially involving stage-specific ribosomes), and protein kinase activity. Reciprocal feedback loops (where components of different pathways cross-regulate each other) were observed between kinases and phosphatases, kinases and the translation machinery, and crucially, between kinases and the proteasomal system, with proteasomal inhibition disrupting promastigote differentiation.

We thank the reviewer for the time and implication dedicated to our manuscript.

Further details are organised by order of apparition in the text:

Material and Methods: while the authors are indicating some key parameters, providing the codes and scripts they used throughout the manuscript would improve reproducibility.

We thank the reviewer for this comment and added the URL for the codes to the data availability section.

Why only 2 biological replicates for RNA while the others layers have 3 or 4?

We agree with the other reviewers and have repeated this analysis to have statistically more robust results.

Is the slight but reproducible increase in median coverage observed for chr 1, 2, 3, 4, 6 and 20 stable on longer culture derived promastigotes and sandfly derived promastigotes ?

No, as published in Barja et al Nature EcolEvol 2017 (PMID: 29109466) and Bussotti et al PNAS 2023 (PMID: 36848551), these minor fluctuations are not predicting subsequent aneuploidies in long-term culture nor in sand fly-derived promastigotes. This information has been added to the text.

Is this change of ploidy a culture adaptation representation rather than a life cycle event as the authors discuss later on? (This is probably an optional request that would be nice to include, if the authors have performed the sequencing of such parasites. Otherwise, it should be mentioned in the discussion).

Yes, this is a well-known culture adaptation phenomenon, on which we have published extensively. We added this conclusion and the references to the text.

L333 "Likewise, stage differentiation was not associated with any major gene copy number variation (Figure 1C, Table 2)". The authors are looking here at steady differentiated stages rather than differentiation itself. "Likewise, stage differentiation was.." would be more appropriate.

We corrected this sentence to 'Likewise, differentiation of promastigotes was not associated with any major gene copy number variation at early passage 2'.

L349-355: have the mRNA presenting change in abundance between stages been normalised by their relative DNA abundance ? Said otherwise, can the wave patterns observed at the genome level explain the respective mRNA level ? Can the authors plot in a similar way the enrichment scores in regards to the position on the genome and can the authors indicate if there is a positional enrichment in addition to the functional one they observe ? This may affect the conclusion in L356-358.

As noted above, we did not see any significant read depth changes at DNA level when comparing amastigotes and promastigotes. Thus there is no need to normalize the RNAseq results to DNA read depth. Furthermore, in our comparative transcriptomics analysis, we only consider 2-fold or higher changes in mRNA abundance (which is far beyond the non-significant read depth change we have observed on DNA level). Manual inspection of the enrichment scores with respect to position did not reveal any significant signal (other than revealing some overrepresented tandem gene arrays where all gene copies share the same location and GO term).

L415 "stage-specific expression changes correlate between protein and RNA levels, suggesting that the abundance of these proteins is mainly regulated by mRNA turn-over". Overstatement. Correlation does not suggest causation. "suggesting that the abundance of these proteins could be regulated by mRNA turn-over" would be more appropriate.

We thank the reviewer for this comment and have corrected the statement accordingly.

Figure 3B, could the authors clarify what are the "unique genes" that are on the infinite quadrants? It seems these proteins are identified in one stage and not the other. This implies that the corresponding missing values are missing non-at random (MNAR). Rather than removing those proteins containing NMAR from the differential expression analysis, the authors should probably impute those missing values. Methods of imputation of NMAR and MAR can be found in the literature. Indeed, the level of expression in one stage of those proteins is now missing, while it could strongly affect the conclusions the authors are drawing in figure 4E regarding the proteins targeted for degradation and rescued in presence of the proteasome inhibitor.

We thank the reviewer for this important comment. However, we would like to clarify several key points regarding the treatment of proteins identified in only one condition.

First, the reviewer assumes that proteins identified in one stage but not the other are necessarily missing not-at-random (MNAR). However, this cannot be definitively established, as these missing values could equally be missing completely at random (MCAR). Without additional information, categorizing them specifically as MNAR may be an oversimplification. More importantly, we have concerns about the reliability of imputation methods in this specific context. Algorithms designed to impute MNAR values (such as QRILC) replace absent data using random sampling from arbitrary probability distributions, typically assuming low intensity values. However, when no intensity value has been detected or quantified for a protein in a given condition, imputing an arbitrary low value raises significant concerns about data interpretation. Such imputed values would not reflect actual measurements but rather statistical assumptions that could introduce bias into downstream

analyses. For instance, imputed values could lead to the conclusion that a protein is not differentially abundant, when in reality it is detected in one condition but completely absent in the other. In our view, there are two biologically plausible scenarios: either these proteins are expressed at levels below our detection threshold, or they are genuinely absent (or present at negligible levels) in the corresponding stage. Rather than introducing potentially misleading imputed values, we chose to treat these as genuine stage-specific differences (presence/absence), which results in infinite fold-changes in Figure 3B. Critically, our approach is strongly supported by independent validation through RNA-seq data, which corroborates the differential presence/absence patterns observed at the protein level. Furthermore, our enrichment analyses reveal significant over-representation of specific biological terms among these stagespecific proteins, providing biological coherence to these findings. These converging lines of evidence (proteomics, transcriptomics, and functional enrichment) strengthen our confidence that these represent biologically meaningful differences rather than technical artifacts. Therefore, we believe our conservative approach of treating these as genuine presence/absence differences, validated by orthogonal data, is more appropriate than introducing imputed values based on arbitrary statistical assumptions. To clarify this section, we modified the text as follows: ‘Only expression changes were considered that either showed statistically significant differential abundance at both RNA and protein levels ($p < 0.01$), or showed significant RNA changes ($p < 0.01$) with the corresponding protein being detected in only one of the two stages. These latter proteins are identified by signals that were arbitrarily placed at the upper (detected in ama) or the lower (detected in pro) parts of the graph. Whether these proteins just escape detection due to low expression or are truly not expressed remains to be established.’

L430-435 "These data fit with the GO [...] the ribosome translational activity (34)." This discussion feels out of place and context. It is too speculative and with little support by the data presented at this stage of the manuscript. It should be removed as Figure 3E or could be placed in the discussion and supplementary information.

We agree with the reviewer. In response to a comment from reviewer 1, we have moved both panels to Figure 2, which much better integrates these data.

The authors present an elegant way to show stage specific degradation through the comparison of stage specific proteasome blockages that show rescue in ama of proteins present in pro and vice versa. L494 "reveal an unexpected but substantial" the term unexpected is inappropriate, as several studies have shown in kinetoplastids the essential role of protein turnover through degradation / autophagy during differentiation. Furthermore the conclusions may be strongly affected by the level of expression of the proteins in the infinite quadrants as we discussed above, and should be revised accordingly.

We rephrased the conclusion to ‘In conclusion, our results confirm the important role of protein degradation in regulating the *L. donovani* amastigote and promastigote proteomes and identify protein kinases as key targets of stage-specific proteasomal activities.’ Please see the response to comment 9 regarding the unique proteins.

L518 "These data reveal a surprising level of stage-specific phosphorylation in promastigotes, which may reflect their increased biosynthetic and proliferative activities compared to amastigotes." Overstatement. Could also be due to culture adaptation - What is the overlap of stage-specific phosphorylations with previous published datasets in other species of Leishmania? Looking at such comparisons could help to decipher the role of culture adaptation response, species specificity and true differentiation conserved mechanisms.

We agree with the reviewer and have toned this statement down by adding the statement ‘...or simply be a consequence of culture adaptation’.

The discussion is extremely speculative. While some speculation at this stage is acceptable, claiming direct link and feedback without further validation is probably far too stretched. For example, the changes of phosphorylation observed on particular sets of proteins, such as phosphatase and DUBs, need to be validated for their respective change of protein activity in the direction that fits the model of the authors. Those discussions should be toned down.

We agree with the reviewer and have strongly toned down the entire discussion, emphasizing the hypothesis-building character of our results, which provide a novel framework for future experimental analyses.

A couple of typos:

In the phosphoproteome analysis section, "...0,2 % DCA..." should be "...0.2 % DCA..." (use a decimal point).

L225 "...peptide match was disable." should be "...peptide match was disabled."

Both corrected

Reviewer #4 (Significance):

While there is not too much novelty around the emphasis of gene expression at post-translational level in kinetoplastid organisms, the scale of the work presented here, looking at 5 layers of potential regulations, is. Therefore, this study represents a substantial amount of work and provides interesting and comprehensive datasets useful for the parasitology community.

We thank the reviewer for this positive statement.

Several potential concerns regarding the biological meaning of the findings were identified. These include the limitations of in vitro systems promastigote differentiation potentially limiting the conclusions, the challenge of inferring causality from correlative "omics" data, and the complexities of functional interpretation of changes in phosphorylation and metabolite levels. The proposed feedback loops and functional roles of specific molecules would require further experimental validation to confirm their biological relevance in the natural life cycle of Leishmania, but that would probably fall out of the scope of this manuscript.

We agree with the reviewer and have modified our manuscript throughout to remove any causal relationships. Indeed, this work is setting the stage for future investigations on dissecting some of the suggested regulatory mechanisms.

Area of expertise of the reviewers: Kinetoplastid, Differentiation, Signalling, Omics

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